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4 April 1974
SUPERSEDING
MIL-C-52326A
23 June 1971

MILITARY SPECIFICATION

CRANE, CRAWLER MOUNTED: 60-TON AT 25-FOOT RADIUS, DIESEL-ENGINE-DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a diesel-powered crawler-mounted crane suitable for use with a clamshell, dragline, and wrecking ball. The crane shall be one size only with a rated lifting capacity of 60 tons at a 25-foot-load radius.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 3810

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SPECIFICATIONS

Federal

O-A-548	- Antifreeze, Ethylene Glycol, Inhibited.
RR-W-410	- Wire Rope and Strand.
VV-F-800	- Fuel Oil, Diesel.
ZZ-H-428	- Hose, Rubber: Preformed and Straight; Hose, Preformed: Flexible, Wire-Reinforced, (for the Coolant Systems of Automotive and Other Liquid-Cooled Engines).
GGG-B-485	- Block, Tackle, Wire Rope, with Swivel Hook; (for Crane Use).

Military

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant, (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-H-3199	- Heater, Vehicular Compartment 28 Volt, DC.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-C-3580	- Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full Revolving, and Their Attachments Packaging of.
MIL-G-3787	- Glass Laminated, Flat; (Except Aircraft).

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MIL-G-3859	- Grease Guns, Hand Operated; Lever, Push and Screw Type.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missiles and Ordnance.
MIL-L-6085	- Lubricating Oil: Instrument, Aircraft Low Volatility.
MIL-H-8796	- Hose, Airduct, Flexible, Aircraft.
MIL-L-10295	- Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-L-10324	- Lubricating Oil, Gear, Sub-Zero.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-B-11040	- Belt, V; Engine Accessory Drive.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-H-13071	- Housing, Lamp, Resilient Mounting, for Construction Machinery.
MIL-C-13516	- Circuit Breakers; Manual and Automatic (28-Volt).
MIL-H-13531	- Hose, Rubber and Hose Assembly, Rubber (Hydraulic, Flexible).
MIL-S-13735	- Switches, Toggle (30 Volts Direct Current Maximum).
MIL-H-13910	- Hydraulic Fluid, Polar Type Automotive Brake, All-Weather.
MIL-C-14055	- Cup, Hydraulic Brake Actuating Cylinder: Synthetic Rubber (General Specification for).
MIL-R-14328	- Rubber Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (for Extreme Climatic Conditions).
MIL-B-16028	- Buckets, Clamshell.
MIL-B-16029	- Buckets, Dragline.
MIL-B-23276	- Balls, Wrecking.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-T-52198	- Taglines, Crane and Crane-Shovel, Spring-Loaded Reel Type.
MIL-M-52590	- Mufflers, Exhaust, Internal Combustion Engines, General Purpose.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.

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STANDARDS

Military

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Heavy Military Equipment.
MIL-STD-421	- Chain, Roller; Power Transmission and Conveyor, Flat Link Plates; Base Pitch, Single and Multiple Strand; Connecting Links and Attachment Links.
MIL-STD-424	- Chain, Roller; Power Transmission, Offset Sidebars, Single-Pitch.
MIL-STD-425	- Chain, Silent; 60 Degree Inverted Tooth, Power Transmission.
MIL- STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1400	- Engines, Gasoline and Diesel, Methods of Test.
MIL-STD-1474	- Noise Limit for Army Materiel.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS35340	- Washer, Lock-Spring, Helical, Extra Duty (Extra Heavy) Series.
MS35647	- Padlocks, Key Operated.
MS35829	- Hinge, Continuous (Piano), .075 Thickness, .187 and .250 Pin Dia.
MS35844	- Plug, Machine Thread, Magnetic: Drain.

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MS49006	-Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.
MS51967	-Nut, Plain, Hexagon-Carbon Steel, Cadmium Plated, UNC-2B (In./MM).
MS51968	-Nut, Plain, Hexagon - Carbon Steel, Cadmium Plated, UNF-2B (In./MM).
MS75058	-Connector, Receptacle, Electrical-Slave, 2 Contact, 24 Volt, Waterproof.
MS90530	-Switch, Pressure-Warning, SPST, Waterproof, 24 Volt DC.
MS90725	-Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A.
MS90726	-Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNF-2A.
MS90727	-Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Alloy Steel, Grade 8, Cadmium Plated, UNF-2A (In./MM).
MS90728	-Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Alloy Steel, Grade 8, Cadmium Plated, UNC-2A (In./MM).

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANS)

National Electrical Code C1.
Z26.1 - Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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REINHOLD PUBLISHING CORPORATION

Engineering Alloys.

(Application for copies should be addressed to the Reinhold Publishing Corporation, New York, NY 10022.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedure AWS B3.0
Welding Handbook

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

ASSOCIATION OF AMERICAN RAILROADS

Rules Governing the Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF LABOR

Occupational Safety and Health Act (OSHA)
Volume 1, General Industry Standard.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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SOCIETY FOR NONDESTRUCTIVE TESTING

Handbook - Vol. II.

(Application for copies should be addressed to the Society for Nondestructive Testing, 914 Chicago Avenue, Evanston, IL 60202.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, NY 10001.)

3. REQUIREMENTS

3.1 Description. The crane shall consist essentially of crawler mounting, full revolving superstructure, and front-end attachments. Each crane shall be complete and ready for operation. Each crane shall be capable of being converted for use in crane, clamshell, dragline, and wrecking ball operations. The crane shall be constructed so that conversion to any of these uses can be effected without the use of special tools or the necessity of installing additional shafts, drums, or clutches. Additional lagging for dragline operations may be used. The crane, equipped with crane boom, shall be capable of traveling over rough ground. Overall height of the crane with a 60-foot boom in travel position shall not exceed 20 feet. The crane shall be reduced in height to the travel position and from the travel position to lifting crane in not more than 30 minutes for each operation.

3.2 First articles.

3.2.1 Preproduction model. The supplier shall furnish one or more cranes for examination and testing within the time frame specified, to prove prior to starting production, that his production methods and choice of design detail will produce cranes that comply with the requirements of this specification. Examination and tests shall be

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as specified in Section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests.

3.2.1.1 Winterized preproduction model. When winterized cranes are specified (see 6.2) in addition to nonwinterized cranes, two preproduction models shall be furnished for examination and test. One of these preproduction models shall be a nonwinterized crane and one shall be a winterized crane.

3.3 Initial production. When specified, (see 6.2), the supplier shall furnish to the Government one or more cranes for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall comply with all provisions of this specification. Unless protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other. Unless otherwise specified herein, rubber products furnished shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the crane by the Government.

3.4.1 V-belts. V-belts shall conform to MIL-B-11040, and all belts for multiple drives shall be furnished as matched sets.

3.4.2 Hose. Coolant hose and hose assemblies shall conform to ZZ-H-428, type I or II, grade A.

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3.4.3 Common parts. Maximum practicable use shall be made of interchangeable hardware and fastening devices and of a minimum number of types and sizes of bolts, screws, nuts, washers, and similar common (standard) parts.

3.4.4 Threaded parts. All capscrews 1/4 inch nominal size and larger shall be in accordance with MS90725, MS90726, MS90727, or MS90728. All nuts 1/4 inch nominal size and larger shall be in accordance with MS51967 or MS51968. Grade B nuts shall be used with grade 5 capscrews and grade C nuts shall be used with grade 8 capscrews. Lockwashers shall be in accordance with MS35340. All other threaded parts shall be in accordance with Handbook H28 except when otherwise specified in specifications referenced herein. No self-tapping threaded parts shall be used. Provisions shall be made to hold the nuts captive when they are not accessible when using common tools.

3.4.5 Hydraulic cups. Synthetic rubber hydraulic brake cylinder cups for master, wheel, and slave cylinders shall conform to MIL-C-14055.

3.4.6 Keys and key slots. Keys and key slots shall conform to SAE J502 and J503.

3.4.7 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment, and without selection for fit or performance.

3.5 Transportability. The crane shall be capable of disassembly in not more than 96 manhours into major components as necessary so that when loaded for shipment in accordance with the Association of American Railroads publication "Rules Governing the Loading of Commodities on Open Top Cars," the load shall not exceed the dimensions of the Railway Clearance Diagram (see figure 1). The crane shall also be capable of reassembly in not more than 96 manhours. Two cranes may be used in disassembly and reassembly.

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3.5.1 Cab. When necessary for transportability, the cab shall be removable to reduce the size, weight, and shape of the cross section of the crane. The framing shall tie the side panels and main portion of the roof together as one unit. Lifting attachments shall be provided as specified in 3.5.4. The supporting members of the cab shall be fastened to the machinery deck by bolts with captive nuts as fasteners. The supporting members of the cab shall hold the cab securely in place under all operating conditions. All panel joints including free-standing edges of the roof shall be provided with angle stiffeners. All electrical, air, and hydraulic lines that must be disconnected to remove the cab shall be equipped with quick-disconnect, self-sealing couplings at the separation lines of the cab and the machinery deck.

3.5.2 Counterweight. The counterweight shall be made of cast iron in a three-piece sectionalized layer form, each section approximately equal weight and shall be capable of being individually removed or replaced utilizing only those lifting provisions provided on the crane.

3.5.3 Superstructure. The superstructure shall be removable and replaceable.

3.5.4 Lifting attachments. Permanently affixed attachments that enable the crane to be lifted in its normal transport position shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.5.5.

3.5.5 Tiedown attachments. Permanently affixed attachments that permit fastening the crane to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209.

3.5.6 Railroad humping test. The crane shall withstand a railroad humping test of 4, 6, and 8 mph in one direction and 8 mph in the opposite direction without permanent deformation to any of its components.

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3.6 Environmental conditions.

3.6.1 Operating temperature and altitude requirements. The crane shall perform as specified at any elevation from sea level (barometric pressure 29.92 inches of mercury) from plus 125° F. to minus 25° F., to 5,000 feet (barometric pressure 24.89 inches of mercury) and a maximum ambient temperature of 107° F.

3.6.2 Storage temperature requirements. The crane shall not be damaged by storage under the following conditions:

Air temperature
High - 160° F. Time period, 1 hour daily.
Low - Minus 50° F. Time period, 6 hours. No solar radiation.

3.6.3 Rain and wind protection. The crane shall start and operate under any conditions of heavy rain and high wind and shall be tested as specified in 4.6.2.31.

3.7 Safety. All belts, gears, shafts, pulleys, sprockets, spindles, drums, flywheels, chains, and other reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of equipment, including exhaust pipes and other lines which may be subject to high temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust, or electrical equipment. Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. All points requiring lubrication during operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing these operations. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. The crane shall meet the requirements Sec 1910.180 of the Occupational Safety and Health Act.

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3.8 Human factors. The characteristics of the superstructure shall provide for operation by personnel ranging from the small man, arctic clothed, through the large man, arctic clothed, in accordance with SAE J833.

3.8.1 Sound level requirement. The sound level of the crane (excluding horns and warning alarms) shall be in accordance with MIL-STD-1474 category D for steady-state noise when tested in accordance with 4.6.2.26. If the crane can be operated at a level that exceeds 85dB(A) the appropriate discussion of the noise hazard per MIL-STD-1474 (MANUALS) shall include the requirement for hearing protection, the noise level of the item, the distance from the item at which 85dB(A) will always be met, and what operational or system configuration situations will reduce the level to 85dB(A). Hazard signs, if required, shall state, "CAUTION, HEARING PROTECTION REQUIRED".

3.9 Maintainability. All major components and installed attachments shall be accessible for maintenance, repair, and replacement. Covers, safety guards, and plates which must be removed for component adjustment or servicing shall be equipped with removable fastenings. Ease of maintenance provisions shall incorporate features assuming clearances for tool and servicing means at temperatures specified. Dimensions of hand access openings shall be in accordance with SAE J925, arctic mittens. All fuel, lubricant, coolant and liquid reservoirs shall drain to the ground. Each maintenance, assembly, or disassembly operation performed shall be accomplished by not more than two men using a hoist if necessary, common tools, and special tools furnished on the crane. After completion of any maintenance operation, the crane shall operate as specified herein. Each operation shown on the following maintenance operations list shall be accomplished by one man in not more than the time specified using special tools furnished with the crane and common tools:

Maintenance operations list.

(a) Remove, replace, and adjust fan belts - 30 minutes.
(b) Remove and replace generator and adjust generator belt tension - 45 minutes.

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(c) Drain engine lubricating oil, remove and replace oil filter elements, and refill crankcase - 1 hour.
(d) Remove and replace fuel filter elements - 30 minutes.
(e) Disconnect battery cables, remove and replace battery, and reconnect battery cables - 30 minutes.
(f) Drain and refill engine cooling system - 30 minutes.
(g) Lubricate the crane and check the lube level in all gearboxes or reservoir in accordance with the lubrication chart - 1 hour.

3.10 Performance.

3.10.1 Crane capacity. Each crane, with an appropriate length of boom and hook block, shall have rated lifting and tipping loads of not less than those specified in table I. The tipping point is that condition where the crawler rollers on the side opposite the load are lifted from contact with the treads when lifting over the side or where one-third of the bearing area of correctly adjusted treads are lifted from the ground when lifting over front or rear. The boom radius is the horizontal distance between the centerline of rotation of the crane superstructure and the centerline of the hook block.

Table I. Crane Lifting Capacities.

Radius of load (feet)	Lifting load (pounds)	Tipping load (pounds)
15	220,000	294,000*
20	160,000	214,000
25	120,000	160,000
30	92,000	123,000
40	60,000	80,000
50	45,000	60,000
60	35,000	47,000
100	15,800	22,000
160	4,250	5,700

* Capacity load. Tipping at 15 feet load radius not required.

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3.10.1.1 Crane-capacity plate. A crane-capacity plate conforming to 3.26 shall be furnished and mounted in a location conspicuous to the operator by the supplier, and shall contain the following information:

(a) Rated loads in 10-foot increments for the full range of operating radii when using a 60-foot boom and for each 20-foot increment in boom length up to and including 180 feet. (The rated load shall be not more than 75 percent of the tipping load.)
(b) Indication that the rated loads are gross hook loads at a radius measured from the center of rotation of the revolving superstructure and that the weights of the bucket or any other external auxiliary lifting means are to be included as part of the hook load.
(c) Statement that the rated loads are based on the crane standing on a supporting surface that is firm, level, and uniform.
(d) Statement of deduction to be made from the rated loads for components of crane attachments such as jibs.
(e) A statement that maximum allowable load is limited by factors other than tipping, if appropriate.

3.10.2 Operating characteristics. All operational functions for a given front-end attachment shall operate independently and simultaneously where required for maximum performance. These functions shall include hoisting and swing simultaneously.

3.10.2.1 Maneuverability. The crane with 60-foot boom shall be capable of traveling both in a forward or in a backward direction, with the revolving superstructure in any position, and shall be capable of being steered either to the right or to the left with controls from the operator's position. The completely equipped crane shall be capable of ascending and descending a 20 percent grade on smooth, firm, dry ground at not less than 0.40 miles per hour, with either a clamshell or dragline bucket located either to the front or the back of the crane. Ground clearance shall

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be at least 12 inches. The crane shall be capable of fording and operation on hard-bottomed bodies of water up to 12 inches for periods of 10 hours duration without evidence of damage or water leakage when tested as specified in 4.6.2.32. The completely equipped crane shall be capable of negotiating a 10 percent side slope on smooth, firm, dry ground.

3.10.2.2 Ground bearing pressure. The ground bearing pressure shall be not greater than 14 pounds per square inch (psi).

3.10.3 Load hoist drum power. Drum power shall be not less than 6,000,000 foot-pounds per minute when operating at full-load governed engine speed, at an actual or simulated altitude of 5,000 feet and a single-line hoist speed not greater than 200 feet per minute.

3.10.4 Backward stability. The center of gravity from the axis of rotation shall be not farther than 70 percent of the radial distance from the axis of rotation to the tipping fulcrum in the least stable direction when the crane shovel is resting on a firm, level, and uniformly even supporting surface; in working order without load; and with the base rating length of crane boom at the minimum working radius.

3.11 Construction details.

3.11.1 Unit stresses. The maximum unit stress imposed by the required loading shall not exceed 60 percent of the yield point of any structural member.

3.12 Superstructure. The superstructure shall consist essentially of a cab enclosing the machinery and operator's position and shall be mounted on a full revolving platform. All operating machinery shall be mounted on and attached to the rotating platform. The boom and machinery shall be attached to the principal members of the platform frame. Walking surfaces and steps, inside and outside the cab, shall be of the antiskid type, and those outside shall be of expanded steel or grillwork which will not accumulate mud or snow.

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Grab handles shall be provided alongside the doors, walks, and steps, and shall be located so as to facilitate safe access to the operator's seat and to components requiring frequent attention. Grab handles shall provide for hand clearance. A catwalk shall be installed all around the superstructure except for the boom foot area. The catwalk shall be not less than 15 inches wide and shall be surfaced with open skidproof grating. The catwalk shall support a concentrated load of not less than 400 pounds per square foot. The catwalk shall be bolted to the superstructure and shall be removable. Access footholds or handholds to enable the operator to gain access to all platforms and catwalks shall be provided. A ladder shall be attached to the front of the superstructure cab to facilitate accessibility to gantry and A-frame sheaves.

3.12.1 Cab. The cab shall be of sheet metal not less than 0.1046 inch in nominal thickness and shall have sufficient windows and glazed doors to permit 180-degree vision for the operator. All glass shall be 1/4-inch nominal thickness, safety type, conforming to ANS Z26.1, type AS2, and held in place by rubber channels for replacement. The front window shall be arranged for 75 percent opening without the use of tools and shall slide into the roof of the cab for storage. All cab doors, except for winterized units (see 3.19.6) shall be the sliding type and shall be restrained from accidental opening and closing while the crane is either traveling or operating. A hasp for a padlock shall be provided. An operator's seat shall be provided. The seat shall have a foam-rubber-padded cushion and back mounted on the same frame. The seat covering shall be vinyl-coated cotton duck material weighing not less than 32 ounces per square yard. The seat shall provide recommended dimensions and ranges of adjustment specified in SAE J899. The crane cab shall contain a heater conforming to MIL-H-3199, type II, for the applicable voltage system. The heater ducts shall be arranged to defrost each window area and shall be mounted and positioned to permit routine maintenance and to allow for quick removal. The cab shall be fabricated so as to be waterproof at all seams and door and window frames. To keep out rain, all joints or seams shall be overlapped or provided with labyrinth-type seals as shown in figures 2 and 3. One or more heavy duty window wiper(s), air or electrically operated shall be

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frunished. The minimum blade length shall be 16 inches. Provision shall be provided for manually parking the wiper when window is stored. The minimum torque rating shall be 50 inch pounds if electric and 100 inch pounds if air operated. If electric operated the switch shall conform to SAE J112.

3.12.2 Gantry. The crane shall be provided with a gantry capable of withstanding stresses induced by lifting a 180-foot boom from the ground to operating position. The principal compression and tension members shall be connected so that the operating stresses are transferred to the main structural members of the rotating platform. The gantry shall be capable of being folded or removed to conform to 3.1.

3.12.3 Powerload lowering device. The primary load hoist drum shall be provided with a reversing drive which will permit lowering the load under the retarding effect of the brakes set to hold the load when tested in accordance with 4.6.2.15. This device shall drive the hoist drum to lower the load at a speed not greater than 50 percent and not less than 30 percent of the drum speed at governed engine speed. The powerload lowering device shall be capable of being engaged by a control located convenient to the operator in his normal seated position within the cab and shall be capable of being disengaged when the crane is used for other purposes which do not require power lowering.

3.12.4 Boom hoist. The crane boom hoist shall be independently controlled. The boom hoist shall safely suspend the boom with recommended reeving without attention from the operator, shall provide for power lowering under the operator's control, and shall make gravity lowering (free spooling of the rope) of the boom impossible under any circumstances. The boom hoist clutch shall be of the friction or electromagnetic type; shall be accessible for adjustment and repair; shall not slip under 150 percent of rated load; and shall be controlled by either mechanical linkage, air, or manual effort applied through hydraulic linkage. The boom hoist shall be equipped with a positive lock to facilitate holding the boom in any set position.

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3.12.5 Drums. The drums shall be independently driven, convertible, suitable for use with any of the different front-end operating equipment, and arranged so as to reduce eccentric pull to a minimum in order to avoid chafing of the rope and unequal wear on the drum shaft bearings. The complete drum assembly and mounting shall rotate freely when clutches and brakes are not engaged to allow gravity lowering of the hook block. The drum shall be equipped with a socket with wedge for dead-ending the drum rope. Lubricating fittings shall be located to permit lubrication without unspooling cable. All drums shall be guarded to prevent the rope from leaving the drum during operation.

3.12.5.1 Drum sizes. Drum lagging shall be furnished with the revolving superstructure to make the unit usable as a crane, clamshell, or dragline. Pitch diameters of the drum laggings shall be in accordance with SAE J881. Drums shall have rope capacity to lift maximum loads for boom lengths of 60 to 180 feet inclusive in 20-foot increments and the required rope lengths (see 3.10.2).

3.12.5.2 Drum brakes. Load brakes, when adjusted, shall hold and lower rated loads on recommended reeving. The brakes shall be capable of being locked in the set position. Brake linings shall be bolted or riveted to the backing plate or band.

3.12.5.3 Drum clutches. Drum clutches shall be of the friction or electromagnetic type; shall withstand high speed and continuous operation without overheating or excessive wear; shall be accessible for adjustment and repair; and shall not slip under 150 percent of rated load. The primary and secondary load hoist drums shall be operated by clutches controlled by levers. The clutch, when engaged, shall be held in place by an over-center lock (knuckle in) on the control linkage. If air controls are used, the valve lever shall latch in place to hold the clutches in engagement when tested as specified in 4.6.2.16. The clutches shall be unlocked by pulling or pushing forward on the control lever. Clutches shall be controlled by either mechanical linkage, air, or manual effort applied through hydraulic linkage or through variable pressure power hydraulic controls. When hydraulic actuators are provided, rotating joints containing a single face type seal which is compatible with military

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fluids shall be provided and shall be capable of operation and storage without leaking or seepage of the hydraulic fluid. A hydraulic tank mounted higher than the highest point in the hydraulic system shall be provided. The clutches shall be normally applied with one application of the control lever or pedal under all conditions specified herein. All clutch facings shall be riveted or bolted to the backing plate. Hydraulic lines shall consist of tubing, hose, and reusable fittings. Hose shall be used where flexing and vibration may contribute to failure. Hose shall conform to MIL-H-13531.

3.12.6 Rollers.

3.12.6.1 Supporting rollers. The rotating frame shall be supported by a live ball, roller ring, or supporting hook rollers. The rollers shall be heat-treated, hardened steel and shall be accurately machined. Fixed rollers shall be fitted with antifriction bearings or replaceable bronze bushings and shall be provided with grease fittings. When the revolving superstructure is tested as specified in 4.6.2.14, the temperature rise of the supporting balls or rollers shall be not more than 175° F. The balls or rollers shall be accessible and shall be removable.

3.12.6.2 Hook rollers. Where the holding-down function is performed by hook rollers, the rollers may be of the dual-purpose type in which one set of rollers performs both the supporting and holding-down functions; or they may be of the holding-down type only. Hook rollers may operate singly or in pairs on tandem bogies and shall be provided with grease fittings. Hook rollers shall be provided with means for vertical or horizontal adjustment. The hookrollers shall be removeable without removal of the hook-roller brackets from the superstructure.

3.12.7 Swing. The swing speed of the superstructure shall be between 2.0 and 3.8 rpm.

3.12.7.1 Swing mechanism. The swing gear pinion shall have accurately machined or ground teeth of alloy steel. The swing gear pinion shall operate in antifriction or bronze bearings equipped with means for lubrication. The rotation of the rotating frame shall be controlled through friction or electromagnetic type clutches designed for fast

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cycling, craning, and other operations, and with a clutch capable of transmitting not less than 130 percent of the maximum engine torque. Clutches shall be controlled by either mechanical linkage, air, or manual effort applied through hydraulic linkage, or by full power hydraulics. Service swing clutches shall make not less than 2-1/2 passes per minute for 1 hour without either fading or exceeding a temperature of 600° F. with a dragline bucket loaded to a gross weight of not less than 18,000 pounds or an equivalent weight. A 60-foot boom shall be used at no higher than a 50-degree angle. A pass consists of a complete swing cycle of 90 degrees one way and 90 degrees back.

3.12.7.2 Swing lock and brake. A swing lock shall be provided to lock the superstructure in any desired position. A swing brake shall also be provided to control the swing of the revolving superstructure while it is operating. Controls for the swing lock and the brake shall be within reach of the operator while seated in the cab.

3.12.8 Power transmission. The engine shall be connected to the transmission through a friction-disengaging clutch and a 3-stage hydraulic torque converter. When the converter is stalled at wide-open engine throttle for 60 seconds, the converter oil temperature shall not exceed 250° F. and shall return to a stabilized temperature within 15 minutes. The torque converter shall be furnished with an overspeed lockup clutch so that converter speed is no faster than the engine speed. The engine clutch shall be controlled from the operator's position, shall be arranged to engage and disengage instantly while the engine is running, and shall require manual effort to engage. The main clutch shall be capable of transmitting not less than 150 percent of the maximum engine torque and shall be located so as to be easily accessible for repair. Drum clutches shall withstand high-speed, continuous operation without overheating or excessive wear. Swing clutches shall be of the friction or electromagnetic type constructed for continuous slippage operation. The main hoist clutches and travel clutch shall each individually be capable of stalling the torque converter and shall be located so as to be accessible for adjustment and repair. Clutches shall

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be controlled by either mechanical linkage, air, or manual effort applied through hydraulic linkage.

3.12.9 Electrical system. A waterproof, 24-volt electrical system for the revolving superstructure shall be furnished. A protective relay assembly shall be provided to protect the complete crane electrical system, especially the alternator, from damage when (1) the batteries are incorrectly installed, (2) incorrect polarity of slave or jumper cables is applied to the slave receptacle, and (3) performing maintenance repairs anywhere on the crane using commercial electrical arc welder. Fuses are not acceptable. When protective diodes are furnished, the diodes shall have a peak inverse voltage (PIV) rating of not less than 1200 volts and a minimum amperage rating of not less than 150 percent of the maximum ampere draw.

3.12.9.1 Floodlights and associated electrical systems. Floodlights shall consist of light housings and sealed-beam units conforming to MIL-H-13071, except as otherwise specified herein. The sealed-beam units shall be 24-volt, 60-watt, floodlamp type, trade number 4578.

3.12.9.2 Wiring harness. The electrical cable shall conform to SAE J558, size number 14 American Wire Gage (AWG) or larger, conforming to the National Electrical Code C1 of the American National Standards Institute and shall be formed into a harness by wrapping with plastic electrical insulation tape. The harness shall be clipped to the frame members or installed in conduit. All lengths of wire or groups of wire exceeding 3 feet shall be placed in conduits. The circuits shall be color coded in accordance with SAE J821 or number traced. Means to prevent chafing of the cable when passing through structural members or sheet metal shall be provided. A terminal block, adequately fused, shall be installed with six terminals for floodlights.

3.12.9.3 Floodlights. Floodlights for illuminating the working area shall conform to trade number 4578, size PAR-46. Each lamp shall be provided with a housing conforming to MIL-H-13071. The housings shall be mounted in a position to permit replacement of the lamp. They shall not extend beyond the front or side of the

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cab of the revolving superstructure unless provision is made for an adjustable bracket to fold the lights inside the transport cube of the crane. The floodlights shall be located so that lenses will not be seen by the operator in his normal operating position and shall illuminate the working area directly in front of the operator.

3.12.9.3.1 Top-mounted floodlights. Four floodlights, two on each side of the boom-hoist cables, shall be mounted at the top front edge of the cab.

3.12.9.3.2 Skirt-mounted floodlights. Two floodlights, one on each side of the superstructure skirt, shall be mounted as near the sides and bottom of the skirt as practicable but not higher than midpoint between the bottom of the skirt and the bottom of the operator's window.

3.12.9.4 Interior lights. The superstructure shall have three ceiling lights effectively located with plastic lens not less than 4 inches in diameter with commercial bayonet type fixture for not less than 21-candlepower lamps (trade number 1204).

3.12.9.5 Trouble light. A handgrip-type, portable trouble light shall be mounted in the cab. The light shall be equipped with trade number 4505, sealed-beam spotlight and supplied with a connecting, self-coiling cable not less than 25 feet long. The trouble light shall have a handgrip handle incorporating a thumb or finger "on-off" switch. The trouble light mounting bracket shall be of the cushioned spring-over-center type, holding and enclosing the sealed beam unit so that it is protected from damage during all operations specified herein.

3.12.9.6 Instrument lights. Instrument lights shall be provided for the illumination of all instruments on each instrument panel. The lights shall provide a lighting intensity of 25 foot-candles on each instrument.

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3.12.9.7 Circuits, circuit breakers, and switches. Lights shall be controlled by single-pole, single-throw type, toggle switches conforming to MIL-S-13735, type I, class B, grade 1; and protected by automatic reset type circuit breakers conforming to MIL-C-13516, type II, grade A, class 1, 15 amperes. Switches shall be located on the instrument panel in the revolving supersturcture within easy reach of the operator when seated. One switch and associated electrical circuit shall be provided for the independent control of the following light combinations, as furnished:

(a) Top-mounted floodlights.
(b) Skirt-mounted floodlights.
(c) Interior cab lights.
(d) Instrument lights.

The starter button, manual engine controls, and gages for the engine shall also be mounted on the above panel with the light switches. An identification plate as specified in 3.27 shall be provided under each switch, control, or gage on the instrument panel. Arrows or any appropriate mark showing the direction the control shall be moved to operate, if pertinent, shall also be placed on the plate.

3.12.9.8 Horn. A horn shall be mounted on the superstructure within the transport cube formed by the crawler and superstructure and shall be slanted downward to drain water. The horn shall be operated by a pushbutton (normally open) switch mounted on the swing level in the cab.

3.12.9.9 Batteries. The batteries shall be in sufficient paired quantity to furnish the required ampere-hour capacity. Batteries shall conform to MS35000-3 except that electrolyte shall be furnished. The batteries shall be shipped charged and dry with a filling of electrolyte separately packaged in accordance with 5.1 for each crane. All lug terminals shall be removable without requiring disassembly of other components. All terminals shall be coated with a phenolic coating for waterproofing.

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3.12.9.9.1 Battery enclosures. Weathertight enclosure of the box-type or compartment-type shall be provided to contain and to protect the batteries for the electrical system. The enclosure shall: (1) have not less than 36 inches ground clearnace; (2) hold the batteries in a horizontal plane when the crane is on level ground; (3) include drains for removal of water or electrolyte spilled within the container, and gas venting at or near the top of the container; and (4) provide accessibility for servicing and for removal of the batteries. The battery enclosure shall provide clear vision of electrolyte level in all cells of the batteries. Sheet metal used to fabricate the enclosure shall be steel having a nominal thickness not less than 0.060 inch. The enclosure shall be treated with a flexible, protective phenolic coating. Each pair of batteries shall be fastened to the bottom of the enclosure or the battery carrier. The box-type enclosure shall be equipped with a piano hinged top cover and the enclosure shall be rigidly fastened to the chassis. A sheet of 1/8-inch-thick rubber or nonconductive material shall be cemented to the bottom side of the horizontal surface of the boxtop cover. The battery box shall have not less than 1-inch clearance on each of the four sides of the 6 TN battery cases as dimensioned in MS35000. On the side opposite the hinge, two adjustable hook latches shall be installed, one near each corner. The cover shall open to an angle greater than 100 degrees to permit vertical removal of the batteries and shall be held in the open position by spring-loaded, detent braces which require manual release. The inside of the cover in the closed position shall have clearance in excess of 1 inch between any part of the cover and battery lug terminals, battery posts, and other battery components. The cover shall overlap the sides and ends of the box.

3.12.10 Electrical generator system. The crane shall be provided with an ac generator system (alternator) in accordance with SAE J180. The heavy-duty charging system shall be the low cut-in type that provides not less than 25 amperes hot charging current at engine idle speed. Mounting shall be in accordance with SAE J545. The alternator speed range shall be in accordance with the recommendations of the alternator manufacturer and shall not exceed 6,000 rpm, maximum.

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3.12.11 Emergency starting and charging receptacle. A slave receptacle conforming to MS75058 shall be provided to facilitate electrical connections between the battery and the external electrical power source for emergency starting of the engine and for charging the batteries. The receptacle shall be recessed in a box in the superstructure cab and shall be accessible to a man standing on the ground. The receptacle or box shall not extend beyond the outside of the cab.

3.12.12 Instrument panel. An instrument panel containing all switches, gages and instruments, except hourmeter-tachometer, for complete control of the engine and lights shall be located within reach and visibility of the operator in his normal operating position. All switches, gages and instruments shall be identified as to nomenclature and positions, as a minimum requirement. Identification plates shall conform to 3.27. All instruments and indicators shall be installed assuring a metal-to-metal return ground from the component to the battery. The instrument cluster group shall be removable from the front to permit access to all wiring and units. Panel lights shall be furnished for visual reading of all instruments, indicators and identification plates by the operator. The instruments and gages shall be mounted within gage functioning tolerance. All warning and indicator lights shall be banked horizontally on the instrument panel. All primary instruments, indicators and controls for diesel-engines as defined in SAE J209 shall be furnished and shall be not less than the following:

Item

Battery generator indicator
Fuel gage
Heat indicator (enging coolant)
  Indicator
  Transmitter
  High temperature warning light
Heat indicator (torque converter and engine)
  Indicator
  Transmitter
  High temperature warning light

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Oil pressure gages
  Indicator
  Transmitter
  Low-pressure warning light
Hourmeter-tachometer

The hourmeter-tachometer may be mounted on the engine in a location that is visible to the operator in his normal operating position. The hourmeter shall have a totalizing mechanism to show not less than 9,999 hours to register the number of hours of engine operation. The tachometer shall be accurate within plus or minus 5 percent of actual engine speed. All panel wiring shall be routed to eliminate a tangle of crossed wiring behind the instrument panel. Electrical wires shall be bundled into a harness assembly routed along an edge of the instrument panel with component connection lines parellel to each other line.

3.12.13 Operator controls. The operator's controls shall be located and installed within the maximum areas in accordance with SAE J898. The crane, and other operating attachments, shall be controlled by the least possible number of hand levers and foot pedals necessary for efficient operation. The levers and pedals shall be within reach of the operator when seated and shall not obstruct the operator's view of the work. All brakes shall permit locking and unlocking by foot action alone but shall not become locked without the operator's intent regardless of the operator's size, skill, or training. The foot-operated brake pedals shall lock and unlock by foot action alone. For ease of operation, the pedals shall rise from the depressed position in a backward as well as vertical movement. Hand and foot operated controls shall be in accordance with SAE J983. Latches on control levers shall not cause delay in operation. An instruction plate conforming to 3.27 shall be located on or near each control level or pedal to indicate the function. The neutral and operating positions shall be shown by arrows and words. When the same lever controls more than one function, all functions shall be listed. The method of lettering and the method of fastening the plate to the panelboard shall be in accordance with MIL-P-514.

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3.12.13.1 Engine hand controls. Mechanical push-pull remote controls, if used, shall be of the waterproof, antifriction type. As a minimum, all exposed parts shall be fabricated of corrosive-resistant steel. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding in the mounts or cable. Bend radii shall be as recommended by the cable manufacturer plus an increase in radii of 20 percent. The as-installed mechanical efficiency of each installation shall be not less than 71 percent and the no-load friction shall not exceed 5 ounces per foot. The continuous operation load rating, in both tension and compression, shall be not less than 500-pound rating for crane controls and 200-pound rating for engine controls and other controls.

3.12.13.2 Accumulator. The accumulator, when furnished in the hydraulic control system, shall be of the piston type and charged with nitrogen gas. The pressure in the accumulator after completion of the tests specified in 4.6.2 shall be not less than it was at the start of the test. The accumulator, when part of a hydraulic control system incorporating an engine driven pump, shall be of sufficient capacity to assure control of a clutch for not less than 15 full cycles of engagement and release without engine power.

3.13 Engine. The commercial diesel engine shall be a heavy duty industrial type, capable of operating on diesel fuel conforming to VV-F-800 and lube oil conforming to MIL-L-2104. The average oil consumption during preproduction and IPT testing shall not exceed .0035 lbs. per brake-horsepower hour. Only full-flow type oil filters shall be utilized in the engine lubricating system. The temperature of the oil in the engine oil gallery shall not exceed 230° F when tested as specified herein. When an oil cooler is provided, the temperature of the oil in the engine sump, or entering the cooler, shall not exceed 250° F when tested as specified herein. The temperature of the coolant in the top tank of the radiator shall not exceed 210° F when tested as specified herein. The coolant temperature, when operating at an ambient temperature of minus 25° F, shall not be less than plus 140° F after a 15 minute warm-up, at not more than one-fourth of the maximum intermittent load rating. The engine shall be the same engine that has been used in the manufacturer's commercial crane of the same size and capacity for at least one year.

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3.13.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the crane under any of the operating conditions specified herein, will not require horsepower in excess of the intermittent rating of the engine as published by the engine manufacturer, or 90 percent of the engine manufacturer's published maximum horsepower rating, at the applied speed; whichever is less. The engine, including accessories, shall be capable of developing intermittent brake horsepower at full-load governed speed of not less than 150 percent of the drum power specified in 3.10.3 at an actual or simulated altitude of 5,000 feet.

3.13.2 Operation conditions. All systems, including lubrication system, induction system, and coolant system, shall perform without leakage or malfunction when the crane is negotiating a 20 percent grade or a 10 percent side slope when tested as specified in 4.6.2.17.

3.13.3 Fuel system. A fuel tank conforming to SAE J703 shall be furnished. The capacity of the tank shall be sufficient for not less than 8 hours of continuous operation under any operating conditions specified herein. Fuel capacity shall be based upon 85 percent of the fuel consumption rate obtained during conduct of testing specified in MIL-STD-1400, test method 2000, at the applicable horsepower at governed speed. The tank shall be located under the machinery deck or at the rear of the cab and just forward of the counterweight. The filler neck of the fuel tank shall extend to a point outside the cab in an accessible and protected location. The tanks shall be of welded aluminum construction. The strainer shall be not less than 3 inches ID and shall be made of bronze screening not less than 60 mesh, protected in such a manner as to prevent contact with the filling nozzle. The fill pipe and strainer shall permit filling the tank at a rate of 50 gallons per minute without spilling. The tank shall be furnished with a drain and threaded plug. The drain shall be located so that removal of the plug will result in complete drainage of the tank directly to the ground. A shutoff valve shall be furnished to control drainage rate when drain plug is removed.

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3.13.4 Starting-priming systems. Unless otherwise specified (see 6.2) engines shall be provided with cold starting priming system either conforming to MIL-E-52649, type III, size as recommended by the engine manufacturer, or of the glow plug type, suitable for cold temperature engine starting for meeting the requirements specified in 3.6.1.

3.13.5 Engine safety devices. Safety devices for low oil pressure and high coolant temperature shall be provided. The safety devices shall operate warning lights instead of stopping the engine. An over-speed safety device shall be provided when positive shut-down of the engine cannot be accomplished from the operator's position. The engine fan shall have a fan guard.

3.13.6 Governor. A governor shall be supplied for the engine limiting the maximum operating speed to a speed required by supplier.

3.13.7 Housing. The engine shall be housed within the revolving superstructure cab.

3.13.8 Exhaust system. The exhaust system shall conform to MIL-M-52590. The exhaust system routing shall expel the gases away from the operator's cab. The exhaust muffler or combination manifold-muffler shall have a volume of not less than 4 times the displacement of the engine.

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3.13.9 Starting systems. The engine starting system shall be as recommended by the engine and the starter manufacturers for this application. The starting systems shall include a starter switch, auxiliary control solenoids, engine speed disconnect, main starter solenoid, starter motor, and flywheel drive mechanism. The starter switch shall be of the waterproof type, rated at not less than 15 ampere inductive load. The starter switch shall actuate a auxiliary control relay which shall actuate the main starter solenoid and starter motor. The engine speed disconnect shall prevent starter engagement whenever the engine is running. The disconnect switch shall conform to MS90530-1 and shall be operated by fuel or oil pressure. The starting system shall withstand at least one discharge of the batteries from full charge (normally 24 volts) to the one-half voltage of not more than 12-volt battery potential under the starter cranking load. Procedure for starter engagement periods shall be in accordance with the starter manufacturer's recommendations.

3.14 Crawler mounting.

3.14.1 Crawler base. The crawler base shall be a casting or shall be rigidly constructed of heavy steel shapes, fitted and welded, and shall be constructed for off-road travel. The swing gear and turntable roller path shall be steel and shall be rigidly attached to or integral with the base. The swing gear shall have accurately machined or ground internal or external teeth. The turntable roller or shear ball path

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shall have an accurately machined or ground bearing surface and shall support the revolving superstructure under all conditions without distortion. The center pin if furnished, shall be of alloy and heat-treated steel and shall have all bearing surfaces accurately machined or ground to size. Where hook rollers perform the holding-down function, the lower member of the center pivot bearing shall be integral with, shrunk into, or welded to the crawler base. The center pin shall operate in bronze sleeve or antifriction bearings.

3.14.2 Track assembly. The crawler tracks shall be of a width and length suitable for the crane; shall be constructed for operation in loose dirt, mud, rocks, and snow; shall be provided with adjustments for maintaining tension in the track belts; and shall prohibit track belts from accidental disengagement. The track shoes shall be of wear-resisting steel and of the flat, self-cleaning type. The track-connecting pins shall be of steel, shall be of sufficient size, hardness, and strength for the purpose intended, and shall be removable and replaceable. The tracks shall be supported on a sufficient number of steel rollers so as to distribute the weight of the entire crane uniformly over the ground for the entire length of the crawlers without depending upon the tension in the track for support. All track-assembly shafts and rollers shall be provided with guarded grease fittings. Where practicable, grease fittings shall be located so as to be accessible from outside the crawlers and lower base.

3.14.3 Side frames. The crawler track side frames shall be steel castings or welded structural steel members. When side frames are not used, large track rollers shall be used with not less than four axles supporting the lower base.

3.14.4 Crawler drive. The crawler treads shall be driven by machine-cut gears or roller chains. The travel mechanism shall have positive steering which shall permit steering while traveling either forward or in reverse, regardless of cab position, by controls within reach of the operator when seated in the cab. The crawler clutches shall be of a size to permit constant operation without overheating. Brakes or other locking means shall be provided to hold the tracks stationary while the crane is pulled across a level, concrete surface,

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and the crane shall be capable of stopping and being parked on the slope specified in 3.10.2.1. Brakes that can be applied at any time regardless of the position of the dogs on the propulsion drive shall be furnished on the travel mechanism. It shall be possible to apply the brakes on both treads at the same time when required. Jaw clutches, when furnished, shall have a sufficient number of teeth to engage and transmit maximum traction power without undue wear. One jaw clutch must be engaged at all times. The travel gears and their bearings shall be enclosed to prevent leakage of lubrication and entrance of dirt or other foreign matter. The propelling shaft shall be arranged so that it is not one or a part of the supporting axles. All shafts shall be bushed or provided with antifriction bearings and ball-check grease fittings.

3.14.5 Axle. Each axle or crossmember, when used, shall be either of cast, forged, or structural steel. When structural steel axles of more than one piece are furnished, the edges to be joined shall be alined and shall have a continuous weld so that the necessary weld penetration for transference of maximum design stress through the base metal juncture shall be provided.

3.15 Front-end attachments. The revolving superstructure of each crane shall be equipped with all necessary equipment to permit operation of the crane as a crane, clamshell, dragline, or with a wrecking ball. When specified (see 6.2) the following attachments shall be furnished with each crane, and such attachments shall be complete with all necessary parts including wire rope:

(a) Boom jib.
(b) Boom, extension, middle, 20 feet.
(c) Clamshell bucket.
(d) Tagline, size 350.
(e) Dragline bucket, 3-1/2-cubic-yard.
(f) Dragline bucket, 3-1/2-cubic-yard, perforated.
(g) Dragline fairlead.
(h) Wrecking ball.

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3.15.1 Crane boom. The crane boom shall include the boom, boom safety stop, boom angle indicator, pendant cables, and necessary harness. Unless otherwise specified (see 6.2), the boom length shall be 60 feet. Boom lengths up to 180 feet shall be accomplished by using 20-foot extensions. Each extension shall be provided with a matched pair of pendant cables. The 60-foot boom shall be made of two or three sections. The boom shall be constructed so that the maximum stress imposed by the required loading shall not exceed 60 percent of the yield point for any structural member when tested as specified in 4.6.2.9. The boom point shall have parallel, guarded sheaves with one more additional sheave than required for lifting maximum loads.

3.15.1.1 Boom safety stop. A telescopic boom stop shall be furnished in order to provide a safeguard against over-topping of the boom. The stop shall be installed so that with a 60-foot boom, a minimum operating radius of 13 feet can be obtained. The complete boom stop with supporting attachments shall have sufficient strength to develop the ultimate strength of the boom in bending at the point of attachment of the boom stop to the boom. The boom stop shall provide a shock absorbing action and shall remain assembled when the boom point is lowered to the ground.

3.15.1.2 Boom hoist cutoff. A boom hoist clutch cutoff shall be furnished which will release the boom hoist clutch and set the brake when a 60-foot boom is raised to a 13-foot operating radius.

3.15.1.3 Boom-angle indicator. The boom-angle indicator shall be of the heavy pendulum type swinging on sealed ball bearings. The pointer shall indicate the angle of the boom with the horizontal plane and shall be visible to the operator. A chart made of the same material as the identification plates and installed in the cab near the operator shall be provided showing rated lifting capacities at various boom angles for various boom lengths with a statement of deductions to be made when using jibs.

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3.15.1.4 Overload indicating system. When specified (see 6.2), an overload indicating system shall be furnished in accordance with SAE J159 and as specified herein. This system shall automatically indicate and warn the operator when an overload condition exists that exceeds the rated loads specified in 3.9.1 through the use of lights, dials, and an audible signal mounted within the operator's view within the cab. The audible and light signals shall cease and dials shall indicate a safe loading condition when corrective action has been taken by the operator. The indicating system may utilize the 24 volt electrical system of the superstructure. The electrical system shall be waterproof and shall operate under all temperatures and use conditions specified herein without any malfunction.

3.15.2 Boom jib. The boom jib shall be 30 feet long and shall be attachable to the crane boom. An overhaul ball, swivel hook with spring loaded safety hook shall be furnished.

3.15.3 Crane hook block. The crane hook block shall conform to GGG-B-485, except for size. The rated capacity of the hook block shall be not less than 110 tons. The hook block shall contain one less sheave than the number of sheaves on the boom point (see 3.14.1). The crane hook block shall be tested as specified in 4.6.2.30.

3.15.4 Clamshell bucket. The clamshell bucket shall conform to MIL-B-16028, type II. The size of the bucket shall be 3-1/2 cubic yards. When specified, (see 6.2) side and corner teeth shall be furnished.

3.15.5 Wrecking ball. The wrecking ball shall weigh not less than 3 tons and shall conform to MIL-B-23276.

3.15.6 Tagline. The tagline shall be of the reel type, with an adjustable tension mechanism, and shall be in accordance with MIL-T-52198, size 350.

3.15.7 Dragline buckets. The dragline buckets shall conform to MIL-B-16029, type II, class S, size 350, or class P, size 350, as applicable.

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3.15.8 Dragline fairlead. The fairlead shall be deck mounted. The fairlead shall be of the 4-sheave, swinging or 2-sheave, revolving type. The fairlead shall have either replaceable wearing plates or replaceable guide rollers; shall have bronze-bushed or antifriction, bearing-mounted sheaves with smooth cable grooves; and shall be arranged to avoid chafing the drag cable in any working position.

3.16 Power-transmission chains. All power-transmission chains shall be provided with a means of adjustment to maintain tension other than by link removal where the center-to-center distance between sprockets is greater than 45 inches. Adjustment of the power-transmission chains bay be accomplished by moving the engine, if the engine and the accessories are mounted on a common base plate. The primary drive chain to the first sprocket shall conform to (a) below and shall be encased by an oiltight steel chain case, and run through an oil bath; or lubricant for the chain may be supplied by a circulating pump directing a spray of oil on top of the top chain span. The transmission chains shall conform to the following:

(a) Flat link plate rol er chains conforming to MIL-STD-421, type IRC, and limited to chain numbers 50-IRC-A, 50-3-IRC, 60-IRC-A, 60-3-IRC, 60-4-IRC, 60-5-IRC, 80-IRC-A, 80-4-IRC, 80-5-IRC, 80-6-IRC, 100-IRC, 100-2-IRC, 100-3-IRC, 100-4-IRC, 120-IRC, 120-2-IRC, 140-IRC, 140-2-IRC, 160-IRC, and 200-IRC.
(b) Offset link plate roller chains selected from table II conforming to MIL-STD-424.
(c) 60-degree inverted tooth silent chains conforming to MIL-STD-425 and limited to chain numbers SC616, SC1020, and SC1024.

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Table II. Chain, Roller:
Offset Link Plates, Power Transmission, Single Strand

Chain No.	1	2	3	4	5	6	7
Nominal pitch (in.)	6.000	5.000	4.500	4.073	3.500	3.067	2.000
Minimum ultimate strength (psi x 105)	3.6	2.5	1.93	1.41	1.22	0.90	0.5625

3.17 Sheaves. The pitch diameter of load bearing sheaves shall be in accordance with SAE J881. Flanges on the boom point sheaves shall extend beyond the rope grooves not less than 1.75 times the diameter of the rope. All sheaves shall be guarded to prevent the rope from leaving the sheave groove during operation. The sheave pins shall be of medium carbon or alloy steel and shall be of sufficient size and strength to support the sheaves under maximum operating conditions.

3.18 Rope. All rope except for the tagline shall be wire rope conforming to RR-W-410, type I, class 2, 6 by 19, construction 3, IWRC improved plow steel uncoated, preformed, regular lay. Wire rope strength factors shall be in accordance with SAE J959.

3.18.1 Rope sockets. Ends of standing ropes shall be secured by zinc-speltered steel sockets, by a flexible loop passing around steel thimbles and spliced into the main rope, or by swage fittings by the rope manufacturer. All other ropes, except tagline lines, shall have ends attached to drum laggings or to dead end connections by wedge-type sockets. Clamps, clips, and screw wedge connectors shall be permitted only on tagline lines.

3.18.2 Rope lengths. Rope shall be of the greatest length required to perform the functions specified in table III using a 60-foot boom. The boom hoist shall have sufficient rope to allow the tip of a 60-foot boom to be lowered to rest on level ground with three wraps of rope remaining on the boom hoist drum.

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Table III. Crane Function to Determine Rope Lengths

Front-end attachment	Position of boom	Position of attachment
Crane hook	At 15-foot radius	At ground level
Clamshell	60 degrees from horizontal	20 percent of boom length (in feet) below ground level
Dragline	40 degrees from horizontal	50 percent of boom length (in feet) below ground level and 30 percent of boom length (in feet) beyond tip.

3.19 Winterization. When specified, (see 3.2.1.1) the crane shall be furnished with a winterization system that will permit efficient engine heating and starting, engine coolant heating, battery heating, personnel cab heating, window defrosting, and equipment operation at any ambient temperature from minus 50° F. to plus 105° F. Cab doors, radiator winterfront and engine housing side panels may be removed for operation above minus 25° F. Engine operation, after starting, shall not be dependent on the winterization system. Winterized cranes shall start in 5 minutes in any ambient warmer than minus 25° F. and function exactly like nonwinterized cranes without overheating or other malfunction at any temperature up to 105° F.

3.19.1 Performance (winterized crane). With the winterized crane stabilized at a temperature of not warmer than minus 50° F., the winterization system shall perform as follows:

(a) Heat the engine, engine coolant, and batteries sufficiently to obtain a start within 5 minutes after 4 hours of standby 115 volt electric heat.

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(b) Raise the engine operating temperature to within the ranges specified in 3.13 in 1 hour after starting.
(c) Heat the personnel cab from minus 50° F to a temperature not colder than plus 35° F. in 1 hour after the engine starts, measured at 6 inches above the floor at the foot pedals, and automatically maintain the temperature of plus 35° F. or warmer.
(d) Defrost not less than 75 percent of each front and side cab window in 1 hour, after the engine starts, and continue to maintain not less than 75 percent of each window frost free.

3.19.2 Engine heater. An electrical standby 115 volt oil pan heater shall be provided with sufficient capacity to raise the engine oil temperature in the pump to an average of plus 60° F. and assure a positive engine start after 4 hours of standby heat. The heater coil shall be installed so it is covered with oil when the oil level is at the low mark and as near the pump pickup as possible. The watt density of the heater shall not exceed 26 watts per square inch. The system shall be controlled by a 180° thermostat. The heater coil shall be covered with oil when the crane is on a 20 percent longitudinal and 10 percent side slope in all directions.

3.19.3 Coolant heater. An electrical standby 115 volt coolant heater, with a thermo-syphon system shall be provided to raise the engine coolant to plus 50° F., measured at the coolant inlet of the heater after 4 hours of standby heat. A thermostat shall be furnished and set to cut off the standby power when the coolant at the inlet of the heater reaches plus 100° F. Shutoff valves shall be installed in the lines to enable removal of the heater without draining the system.

3.19.4 Cab heater(s) and defroster(s). The cab heater(s) shall be a conventional coolant-heat-transfer type, mounted in the cab so as not to interfere with the operator(s) performing their normal functions, or obstruct entering or leaving the cab. A ducting system shall be provided to direct cab heat to the operator's feet and to defrost and maintain not less than 75 percent of each window (front

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and side) frost free. The controls for the heater shall be accessible to the operator and the heater thermostatically controlled to maintain the temperature requirements specified in 3.19.1. The heater shall be provided with an air intake proportioning device with connection and controls to draw any desired mixture of inside and outside air.

3.19.4.1 Coolant lines. The heater lines shall be in accordance with 3.19.12 and shall be of the size specified for the heater. Lines shall be connected so as to insure maximum circulation of the heated coolant throughout the full length of the engine cylinder block and heads. The shortest possible coolant lines shall be used and shall be supported to minimize vibration or interference with engine maintenance and operation. Flexible metallic or flexible steel-reinforced rubber hose and connections shall be used to prevent damage by line vibrations, expansion, or contraction.

3.19.5 Battery winterization.

3.19.5.1 Battery box. The battery box shall be made of corrosion-resistant steel not less than 0.060 inch thickness. The box shall provide a minimum internal clearance of 1 inch around the perimeter and 2 inches over the top of each battery. Batteries shall be supported on nonconductive rails of multiple-post at least 1 inch above a perforated plate over a bottom-plenum at least 1 inch deep. The box shall incorporate a gasketed cover, a drain not less than 1/2-inch ID, provision for firm, acid-resistant battery holddown and spacing, a heating-air inlet to the bottom-plenum not less than 3 inch diameter at least 15 holes in the perforated-plate with a total cross sectional area not less than 10 square inches, and check-dampered 3-inch diameter air outlet vent. The battery supports, holddowns and spacers shall not interfere significantly with the free circulation of heating air through the spaces around the batteries. Rubber grommets or boots shall be provided to cushion and seal the opening where battery cables enter the box. The cover of the battery box shall not be used as a holddown device, shall be provided with latches and handles large enough to facilitate operation by arctic-mittened hands and shall be faced with nonconductive material on the bottom

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side to provide a safety factor against accidental short circuiting of batteries or cables. The battery box shall be so located and installed as to allow full accessibility to the batteries for servicing and replacement and to facilitate connection of a duct from the cab-heating system. If installed in the cab, the box shall be on or near the floor, the drain shall be extended by acid and alkali resistant pipe or hose to discharge on the ground, the outlet vent shall be extended out of the cab to a downward discharging outlet. If not installed in the cab, the battery box and its cover shall be insulated externally with a 1-inch thickness of plastic foam, and the insulation shall be protected by a hard facing on the cover and on exposed sides or ends. Bottom support shall not be through the insulation unless it can be shown that its strength and rigidity are suitable for such application.

3.19.5.2 Battery heating and temperature control. The batteries shall be heated during operations by fresh air from the cab heater, delivered through a duct of at least 3-inch diameter, through a check-damper, and through a thermostatically controlled damper, the latter governed by a temperature sensing element in the battery box to maintain air-space temperature there normally between 40° and 100° F. However, a bypass or override control shall be provided, operable from the panel containing other heater controls, by which air at maximum system-temperature can be delivered continuously when preheating after "soakout" to extreme low temperatures. The positions of this control shall be clearly marked "BATTERY-PRE-HEAT" and "BATTERY-NORMAL-HEAT".

3.19.5.3 Battery charger. A 24-volt, 15 amp charger shall be mounted in the cab to maintain the batteries at the initial charge level during standby operations. The charger shall be equipped with a limiting device to reduce the charger output commensurate with the battery state of charge. A direct reading ammeter indicating the charger output shall be fitted to the charger case.

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3.19.6 Winterized cab. The winterized cab shall be as specified in 3.11.1 and the additional requirements specified herein. The doors shall be of the swinging type and rubber sealing gaskets shall withstand temperatures of minus 50° F. to plus 105° F. Doors shall open a minimum of 90 degrees. Doors shall be equipped with positive latches of a type that will keep the doors closed when subject to vibration. Positive hold open devices to secure doors in the open position shall also be provided. Friction or coil-spring compression devices are not acceptable. All windows except one front window, shall be tinted safety glass conforming to MIL-G-3787, class II, except one ply shall be tinted light green color. All controls or open housings shall be covered with a boot to prevent cold air from entering the cab. A storage compartment shall be provided for the electrical 25-foot cable specified in 3.19.8.

3.19.6.1 Front window. The front window glass shall be laminated, tinted, safety-plate glass conforming to MIL-G-3787, class I, type B. The front window shall be equipped with one or more 24-volt electric window wipers. The wiper(s) shall have not less than two speeds of operation, and shall have a park position for holding the blade when the glass is stored overhead. The wipers shall clear 75 percent of the front window area when in operation and shall be capable of operation from inside the cab by handcrank.

3.19.6.2 Cab insulation. The interior of the cab shall be insulated with mastic-type insulation not less than 3/32 inch thick or polyvinyl-choloride sheet insulation not less than 1/4 inch thick. All insulation shall be nonhygroscopic, fire retardant, resistant to attack by rodents, insects and fungi, and shall maintain its form under all operating and environmental conditions specified herein.

3.19.6.3 Weatherstripping. Weatherstripping conforming to MIL-R-14328 shall be used for sealing the doors and covers to the cab framework and shall not interfere with normal operation of the doors or covers. The type of weatherstripping and method of securing it shall prevent the entrance of blowing snow and shall reduce vibration and noise to a minimum. The weatherstripping used for sealing the bottom of the doors shall be so installed that it will not be damaged by personnel stepping on it while entering or leaving the cab.

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3.19.6.4 Floor mat. The floor of that part of the cab occupied by the operator shall be fitted with a fiber glass or with polyvinylcholoride shell insulated not less than 3/8 inch thick with an integral antiscuff skin surface. The floor mat shall be in sections conforming to the size of the floor plates to permit removal of floor-plate sections without disturbing adjacent areas of the mat. Floor insulation shall be bonded to floor plates.

3.19.7 Radiator winterfront. The radiator shall be furnished with a two piece removable, manually controlled metal winterfront (radiator shutter) of the sliding type. The winterfront shall not jam from blowing snow or become inoperative when covered with a sheet of ice up to and including 1/4 inch thick. The open area shall be at least 30 percent of the frontal area of the radiator core. A locking device shall be provided on each section to positively lock the winterfront in at least the following positions: Full open, half open and closed. Operations of the winterfront shall be accomplished without the use of tools or pry bars.

3.19.8 Wiring system. The electrical 115 volt heating elements for the engine sump, engine coolant and battery box shall be connected through a distribution box with a separate circuit-breaker switch for each outgoing line. This box shall be weatherproof or shall be mounted in the cab, and connected to an easily accessible, externally mounted waterproof male receptacle. A single 3-wire power cable, 25 feet long, conforming to MIL-C-3432, class II, and of adequate capacity to serve all heating elements shall be furnished. Cable terminals shall be of three-prong type and shall ground the vehicle when connected. All winterized elements and the electrical system shall be radio suppressed and shall conform to 3.20.

3.19.9 Air cleaner. Warm air from the engine housing may be directed through the air cleaner if necessary for more efficient engine operation.

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3.19.11 Hose and hose assemblies. Hose and hose assemblies shall conform to the following specifications:

Hot air duct hose.	MIL-H-8796
Coolant hose (radiator and coolant heater).	ZZ-H-428, type I or II, grade A, class 2.

3.19.12 Fuels, lubricants, hydraulic fluids, and antifreeze. The crane shall be assembled, run-in, tested, and delivered with the products listed in table IV, except that fuels and oils used during run-in and testing shall be suitable for use in the prevailing ambient temperature range.

Table IV. Standard Military Fuels, Lubricants, Hydraulic Fluids, and Antifreeze for Winterized Cranes

Product	Specification No.	Symbol
Compound, antifreeze, arctic type.	MIL-A-11755	
Fuel oil, diesel, arctic grade.	VV-F-800	DF-A
Lubricating oil, internal-combustion engine, subzero.	MIL-L-10295	OES
Lubricating oil, gear, subzero.	MIL-L-10324	GOS
Greases, aircraft and instrument.	MIL-G-23827	GL
Greases, automotive and artillery.	MIL-G-10924	GAA
Hydraulic fluid, nonpetroleum arctic type (brake system).	MIL-H-13910	HBA
Hydraulic fluid, petroleum base.	MIL-H-5606	OHA
Lubricating oil, instrument.	MIL-L-6085	OAI

3.19.13 Fabrication. The completely winterized crane shall conform to all the requirements specified herein for nonwinterized cranes including the weight and dimensional requirements. All latches, handles, grabhandles and access doors shall be proportioned to permit grasping and manipulating by a man wearing arctic type mittens.

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3.20 Electromagnetic interference. The electromagnetic interference emission of the crane shall conform to MIL-STD-461, Class III C.

3.21 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition as specified in 3.21.1 with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.21.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66 - Air Pollution Control District, County of Los Angeles, CA.

3.22 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.22.1 Lubricants. The crane shall operate as specified herein when lubricated with military lubricants (see 6.4).

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3.22.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.22.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.22.4 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drainplug conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the crane is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground.

3.22.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.
(b) MIL-G-10924 - For bearings where operating temperatures do not exceed 175° F.

The crane shall be assembled, run-in (broken-in), tested and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

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3.22.6 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range as specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.23 Toolbox. A toolbox made of steel not less than 0.0747 inch in nominal thickness shall be attached to the crane. The toolbox shall be large enough to contain all the tools required for maintenance and operation of the crane. The box lid shall be secured to one edge of a side by a continuous hinge, conforming to MS35829 (pin dia. 0.25 inch, brass), extending the full length of the box edge. The lid shall be held in the closed position by not less than two adjustable-type hook latches with strikers. A hasp shall be located to secure the lid to the box and shall accommodate a lock conforming to MS35647. The toolbox shall be securely mounted in an accessible, protected location permitting the lid to be opened through an arc to not less than 90 degrees. The lid shall be held fully open by the weight of the lid or by a holding device. Drainage of the toolbox shall be provided.

3.24 Antirotation device. An antirotation device shall be furnished for securing the superstructures to the car body of the crane to prevent rotation during shipment. The device shall be of a size to restrain a movement equal to two times the gross weight of the superstructure multiplied by the distance from the center of gravity to the center of rotation. One end of the device shall be permanently affixed in a single position on the car body of the crane or superstructure while in use as an antirotation device and for storage. Means shall be furnished to secure the loose end of the device to the car body or superstructure for storage.

3.25 Identification marking. The crane shall be identified in accordance with MIL-STD-130. The marking shall be applied to the crane on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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3.26 Shipping data plate. Shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and in addition, shall show the silhouette of the crane in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.27 Instruction plates. Each crane shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the crane. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location. The supplier shall install the electrical powerline warning decals furnished by the contracting officer (see 6.5).

3.28 Vehicle marking. Lifting and tiedown attachments shall be identified with 1-inch letters indicating "SLING POINT" and "TIE-DOWN". Unless otherwise specified herein, stenciling shall conform to MIL-T-704. Registration numbers shall be marked on the crane in accordance with MIL-STD-642. The color shall be white.

3.29 Treatment and painting. The portions of the crane normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.30 Servicing and adjustment. The supplier shall service and adjust all cranes for operational use including at least the following: Adjustment of lights; adjustment of the clutch and brake systems and mechanical components of the crane engine, transmission, and drive train; servicing of the cooling system of nonwinterized cranes with a 50-50 solution, by volume, of ethylene-glycol-type antifreeze conforming to O-A-548, type I, and water; and cleaning, bleeding, and filling of the hydraulic systems. The crane shall be completely lubricated as specified in 3.22.

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3.31 Workmanship.

3.31.1 Castings and forgings. All parts, components, and assemblies of the crane which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall not reduce typical mechanical properties or affect a castings function.

3.31.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.31.3 Welding. The surfaces of parts to be welded shall be free from rust scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the crane to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified herein, the welding process used in fabrication of the crane shall be at the option of the supplier.

3.31.4 Welders. Before assigning any welder to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder and

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automatic welder equipment operators have passed qualification tests as prescribed by either of the following listed codes for the materials required to be welded and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME.

Suppliers who only make horizontal welds need not qualify welders for "all position welding".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards, and drawings.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

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4.3 Preproduction inspection.

4.3.1 Examination.

4.3.1.1 Pretest examination. The crane shall be subjected to the examination marked "X" under column 1 of table V. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test examination. Upon successful completion of all tests specified in 4.3.2, the crane shall be subjected to the examinations marked "X" in column 2 of table V. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests.

4.3.2.1 Crane. Upon successful completion of the examination specified in 4.3.1.1, the crane shall be subjected to the tests marked "X" under column 1 of table VI. Failure of any test shall be cause for rejection.

4.3.2.2 Crane components. Crane components listed in 3.2.2 shall be subjected to tests marked "X" under column 1 of table VI. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production cranes will be selected at random by the Government from the cranes being produced by production tooling and will be examined for the defects marked "X" in column 3 and 4 of table V and subjected to the tests marked "X" in column 2 of table VI, to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production crane shall not exclude the remaining cranes from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional

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tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production crane to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production cranes and shall be cause for refusal by the Government to continue acceptance of production cranes until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on cranes previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all cranes accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. After successful completion of all tests specified in 4.5.2, each crane shall be subjected to the examinations marked "X" in column 5 of table V. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each crane shall be subjected to the tests marked "X" under column 3 of table V. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table V.

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Table V. Examination Schedule

Pre-Production		Initial Production					Requirement Paragraph(s)
Pre-test	Post test	Pre-test	Post test	Quality conf.		Defect	
1	2	3	4	5	6		7
X	-	-	-	-	101.	Height not as specified.	3.1
X	X	X	X	X	102.	Parts or components missing	3.1
X	X	X	X	X	103.	Assembly incomplete or incorrect.	3.1
X	-	-	-	-	104.	Material not as specified	3.4
X	-	-	-	-	105.	Cab not removable.	3.5.1
X	-	-	-	-	106.	Quick disconnect couplings not as specified.	3.5.1
X	X	X	X	-	107.	Safety provisions not as specified	3.7
X	X	X	X	-	108.	Human factors not as specified.	3.8
X	-	-	-	-	109.	Superstructure assembly not as specified.	3.12
X	X	-	-	-	110.	The walkways, ladders, steps, and handholds not as specified.	3.12
X	X	-	-	-	111.	Cab vision, glass and rubber channels not as specified.	3.12.1

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Table V. Examination Schedule (Cont'd)

1	2	3	4	5	6	7
X	X	-	-	-	112. Operator's seat not as specified.	3.12.1
X	X	-	-	-	113. Cab heater not as specified.	3.12.1
X	X	X	X	X	114. Windshield wiper not as specified or does not function.	3.12.1
X	X	-	-	-	115. Gantry not as specified.	3.12.2
X	X	-	-	-	116. Boom hoist not as specified.	3.12.4
X	X	-	-	-	117. Drums, drum sizes drum clutches and brakes not as specified.	3.12.5
X	X	-	-	-	118. Rollers not as specified.	3.12.6
X	X	-	-	-	119. Swing mechanism, swing lock and brake not as specified.	3.12.7
X	X	-	-	-	120. Torque converter and clutch not as specified.	3.12.8
X	X	X	X	X	121. Electrical system not as specified or does not function.	3.12.9

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Table V. Examination Schedule (Cont'd)

1	2	3	4	5		6	7
X	X	-	-	-	122.	Floodlights and associated systems not as specified.	3.12.9.1, 3.12.9.3
X	X	-	-	-	123.	Wiring harness, circuits, circuit breakers and switches not as specified.	3.12.9.2, 3.12.9.7
X	X	-	-	-	124.	Interior, trouble and instrument lights not as specified.	3.12.9.4, 3.12.9.5, 3.12.9.6
X	X	X	X	X	125.	Horn not as specified or does not operate.	3.12.9.8
X	X	-	-	-	126.	Batteries and battery enclosure not as specified.	3.12.9.9, 3.12.9.9.1
X	X	-	-	-	127.	Electrical generator system not as specified.	3.12.10
X	X	-	-	-	128.	Emergency starting and charging receptacle not as specified or does not operate.	3.12.11
X	X	-	-	-	129.	Instruments and instrument panel not as specified.	3.12.12
X	X	X	X	X	130.	Operator controls and markings not as specified.	3.12.13

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Table V. Examination Schedule (Cont'd)

1	2	3	4	5	6	7
X	X	-	-	-	131. Accumulator not as specified.	3.12.13.2
X	X	-	-	-	132. Engine and accessories not as specified.	3.13
X	X	-	-	-	133. Fuel system not as specified.	3.13.3
X	X	-	-	-	134. Starting-priming system not as specified.	3.13.4
X	X	-	-	-	135. Engine safety devices not as specified.	3.13.5
X	X	-	-	-	136. Cooling system not as specified.	3.13
X	X	-	-	-	137. Exhaust system not as specified	3.13.8
X	X	-	-	-	138. Crawler base not as specified.	3.14.1
X	X	-	-	-	139. Track assembly not as specified.	3.14.2
X	X	-	-	-	140. Side frames, crawler drive and axle not as specified.	3.14.3, 3.14.4, 3.14.5
X	X	X	-	X	141. Front end attachments not as specified.	3.15

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Table V. Examination Schedule (Cont'd)

1	2	3	4	5		6	7
X	X	-	-	-	142.	Boom safety stop and boom hoist cutoff not as specified.	3.15.1.1
X	X	X	-	X	143.	Boom angle indicator, load chart or overload indicating system not as specified.	3.15.1.3, 3.15.1.4
X	X	-	-	-	144.	Power transmission chains not as specified.	3.16
X	X	-	-	-	145.	Sheaves and sheave guards not as specified.	3.17
X	X	X	X	X	146.	Rope, rope lengths, and sockets not as specified.	3.18, 3.18.1, 3.18.2
X	X	-	-	X	147.	Winterization not as specified.	3.19
X	X	X	X	X	148.	Fungus and moisture resistance not as specified.	3.21
X	X	X	-	-	149.	Lubrication fittings and pressure release device not as specified.	3.22
X	X	-	-	-	150.	Filling, draining and checking provisions not as specified.	3.22.4

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Table V. Examination Schedule (Cont'd)

1	2	3	4	5	6	7
-	X	X	X	X	151. Lifting and tie-down attachments not as specified	3.5.4, 3.5.5
-	X	-	X	X	152. Antirotation device not as specified.	3.24
X	X	X	-	X	153. Identification marking not as specified.	3.25
X	X	X	-	X	154. Shipping data plate not as specified.	3.26
X	X	X	-	X	155. Instruction plates not as specified.	3.27
X	X	X	-	X	156. Vehicle marking not as specified.	3.28
X	X	X	-	X	157. Treatment and painting not as specified.	3.29
X	-	X	-	X	158. Servicing and adjustment not as specified.	3.30
X	X	X	-	-	159. Toolbox not as specified.	3.23
X	X	X	X	X	160. Workmanship not as specified.	3.31

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The crane shall operate as

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specified herein without maintenance other than the supplier's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the supplier prior to test. All lights shall be turned on during all succeeding operational tests. The front-end accessories and parts required to rig the crane for the tests specified herein but not included as part of the contract or order or as Government-loaned property shall be furnished by the supplier at not cost to the Government. Unless otherwise specified herein the test area shall be examined to determine that:

(a) The test is to be performed on a hard surface.
(b) The test area does not exceed plus or minus 1 percent grade in any direction for the lifting capacity and backward stability tests.

The crane shall be checked prior to and during the tests to determine the following:

(c) Insure that coolant and lubrication systems are full.
(d) Check that fuel tanks are maintained between full and halffull level throughout these tests. Check fuel gages for accuracy at empty, one-half full, and full levels.
(e) Check setting of crane engine governor to insure compliance with 3.13.7.
(f) Adjust clutches and brakes in accordance with the manufacturer's instruction manual. Inspection shall be made to insure that there is no dragging of the clutches or brakes with the controls in the disengaged position.
(g) Examine all cable anchors, sheaves, revolving base rollers, drive chains, rotating gears, rotating drums and ropes to insure that there is no rubbing or interference. This examination shall be made at all operating positions of the front-end attachments and shall include the boom raised to maximum height.

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4.6.2.1.1 Pretest accumulator charge pressure. When a hydraulic control system is furnished for the superstructure (see 3.12.5.3) the accumulator charge pressure shall be measured prior to performing any tests.

4.6.2.1.2 Post test accumulator charge pressure. After completion of all tests, when a hydraulic control system for the superstructure is furnished, the accumulator charge pressure shall be measured. An accumulator charge pressure less than that measured in 4.6.2.1.1 shall constitute failure of the test.

4.6.2.2 Test schedule. Tests shall be in accordance with table VI.

Table VI. Test Schedule

Preproduction test	Init. Prod.	Qual. Conf.	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	X	X	Test conditions.	4.6.2.1	3.22
X	X	-	Accumulator pretest.	4.6.2.1.1	3.12.13.2
X	X	-	Accumulator post test.	4.6.2.1.2	3.12.13.2
X	-	-	Engine.	4.6.2.3	3.13
X	-	X	Lubrication.	4.6.2.4	3.22, 3.9
X	-	-	Clearance.	4.6.2.5	3.1
X	-	-	Ground bearing pressure.	4.6.2.6	3.10.2.2
X	-	-	Brake.	4.6.2.7	3.14.4
X	-	-	Stability.	4.6.2.8	3.10.4
X	-	-	Maximum working stress.	4.6.2.9	3.11.1
X	-	-	Ease of maintenance.	4.6.2.10	3.9
X	-	-	Lifting capacity.	4.6.2.11	3.10.1.1
X	-	-	Crane capacity plate determination.	4.6.2.11.1	3.10.1.2

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Table VI. Test Schedule (Cont'd).

1	2	3	4	5	6
X	X	-	Craning.	4.6.2.11.2	3.9.2
X	-	-	Overload indicating system.	4.6.2.11.3	3.15.1.4
X	-	-	Rope strength.	4.6.2.12	3.18
X	-	-	Power and speed test.	4.6.2.13	3.10.3, 3.18.2
X	-	-	Swing test.	4.6.2.14	3.12.6, 3.12.7
X	X	-	Clamshell and dragline.	4.6.2.14.1	3.15, 3.1
X	-	-	Power lowering.	4.6.2.15	3.12.3
X	-	-	Load holding	4.6.2.16	3.12.5.3
X	X	-	Propelling.	4.6.2.17	3.10.2.1
X	-	-	Torque converter stall.	4.6.2.18	3.12.8
X	-	-	Hoist and travel clutch stall.	4.6.2.18.1	3.12.8
X	-	-	Sectionalization.	4.6.2.19	3.5
X	-	-	Tiedown attachment.	4.6.2.20	3.5.5
X	-	-	Lifting attachments.	4.6.2.21	3.5.4
X	-	-	Antirotation.	4.6.2.22	3.24
X	-	-	Electromagnetic interference.	4.6.2.23	3.20
X	-	-	Operating temperature.	4.6.2.24	3.6, 3.13
X	-	-	Winterization.	4.6.2.25	3.19
X	-	-	Sound level.	4.6.2.26	3.8.1
-	-	X	Production run-in.	4.6.2.27 thru 4.6.2.27.3	3.1, 3.30
-	X	-	Maintenance package.	4.6.2.28	3.21
X	-	-	Boom stop.	4.6.2.29	3.15.1.1, 3.15.1.2
X	-	-	Crane hook block.	4.6.2.30	3.15.3
X	-	-	Rain test.	4.6.2.31	3.15.3
X	-	-	Fording.	4.6.2.32	3.10.2.1
-	X	-	Fording.	4.6.2.32(b)	3.10.2.1

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4.6.2.3 Engine. Prior to installation of the engine in the preproduction model, the engine shall be tested in accordance with MIL-STD-1400, test method 2000. Nonconformance to 3.12 shall constitute failure of the test. Upon successful completion of this test and prior to installation of the engine in the preproduction model, the fuel injection pump, governor, and turbocharger, if furnished, shall be sealed by a Government representative at the settings established by this test. Seals for the engine shall also be placed by a Government representative to prevent separation of the crankcase from the engine block and the rocker-arm cover from the engine head. Any changes made to these settings or removal or alteration of the seals by other than the Government representative or without his expressed consent will invalidate all tests previously preformed.

4.6.2.4 Lubrication test. Lubricate the crane as recommended by the manufacturer. Inability to conform to the following shall constitute failure of this test:

(a) That the seals retain lubricant.
(b) That all fittings are accessible without removal of parts or components of the crane.
(c) That fittings comply with 3.22.2.
(d) That fittings are protected from damage due to operation or movement of the crane.
(e) That provision has been made for lubrication of gears, chains, and sprockets.

4.6.2.5 Clearance test. Drive the crane with 60 feet of boom under a line which is located no more than 20 feet above the ground. Any interference between any part of the crane with the line shall constitute failure of this test.

4.6.2.6 Ground bearing pressure test. The ground bearing pressure is derived by dividing the total working weight with complete front-end equipment, but not including load, by the bearing area in square inches. The weight shall be determined by platform scales or by calibrated hydraulic or electronic devices and the bearing area shall be the center-to-center distance between crawler end sprockets multiplied

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by twice the width of the track shoe. Ground bearing pressure over 14 psi shall constitute failure of this test.

4.6.2.7 Brake test. a. Pull the crane with the brakes set on a level (plus or minus 1 degree) concrete surface. Inability of the brakes to hold the tracks stationary while the crane is moved shall constitute failure of this test. b. The crane shall be equipped with the front end attachment to provide the maximum crane weight. With the crane track in forward travel, ascend the grade specified in 4.6.2.17, stopping three times on the grade, and engage and lock the track locking mechanism at each stop. Descend the above grade, stopping three times on the grade and engage and lock the track locking mechanism at each stop. Inability of the track locking mechanism to hold the tracks stationary or evidence of permanent deformation shall constitute failure of this test.

4.6.2.8 Backward stability. With the 60 foot boom at its minimum working radius, specified in table I, in the least stable direction, determine the location of the center of gravity as follows:

$$X = \frac{Rd}{W}$$

where: X = Distance between centerline of crawler track roller opposite lifting side of crane to center of gravity.
R = Weight measured when crawler track opposite lifting side of crane shovel is on firm ground and the crawler track on lifting side of crane shovel is on a platform scale.
d = Distance between centerlines or on center of outer flange of the crawler track rollers.
W = Weight measured when entire crane shovel is on platform scales.

Nonconformance to 3.10.4 shall constitute failure of this test.

4.6.2.9 Maximum working stress test. This test shall be conducted as follows:

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(a) Preparation:
  (1) Crane shall be in working condition with fuel tank at least half-full, with full cooling system, and with full lubricating system.
  (2) All exterior surfaces shall be cleaned of grease, oil, and dirt for strain gage installation.

(b) Installation of strain gages. Strain gages shall be installed in accordance with section 54, Resistance Strain Gage Testing, Nondestructive Testing Handbook. Where directions of principal stress are quastionable, two or more gages or strain-gage rosettes may be employed. During test operation or visual examination, any points which show visual evidence of weakness due to design or paint checking should be gage tested in addition to those points listed in (d). Particular note shall be made of residual stresses in the structural members due to dead loading. These stresses shall be relieved prior to zeroing the strain-recording instrument (i.e., resting the crane boom on a support with cable slacked).

(c) Instrumentation operation. Operation of strain-gage recording or indicating apparatus shall be in accordance with the instrument manufacturer's recommended procedure. Standard commercial instruments specifically designed for resistance strain gages shall be considered acceptable provided the manufacturer guarantees an accuracy to within plus or minus 10 microinches at a scale reading of 600 microinches.

(d) Locations for strain gages shall be determined by photo elastic stress analysis or stresscoat analysis. These locations shall include but not be limited to the following locations:

(e) Location of strain gages:
Superstructure
*Front hook roller bracket.
*Rear hook roller bracket.
Hook roller path - Top side, outer edge, radially.
*Hook roller path - Top side, outer edge, tangential.
Boom foot brackets.

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A-frame compression members - Top midsection.
A-frame compression members - Side midsection.
A-frame tension members - Rear midsection.
A-frame tension members - Side midsection.
A-frame spreaders - Top midsection.
A-frame spreaders - Front midsection.
A-frame head shaft - Top midsection.
A-frame head shaft - Front midsection.
Other critical points on A-frame.
*Drum mounting - At points of critical stress.
*Rotating base - At points of critical stress.
Crane boom
Boom base section - Top of upper chords.
Boom base section - Side of upper chords.
Boom base section - Bottom of lower chords.
Boom base section - Side of lower chords.
Boom tip section - Top of upper chords.
Boom tip section - Side of upper chords.
Boom tip section - Bottom of lower chords.
Boom tip section - Side of lower chords.

* Recommend rosette gages.

(f) Procedures:

(1) General. Repeat strain test for each strain gage for at least three operating test cycles with applicable attachment. Superstructure tests shall be conducted in conjunction with crane boom tests. In addition to the strain gage tests, visual examination shall be made immediately after each test cycle to note any indication of yield (the yield point shall be construed to be any form of permanent deformation to include bending, weld failures, cracks, and scoring of bearings, pins, and bushings). Where rotation of drums, sheaves, or gearing becomes eccentric due to test loading, remove shafts and check for permanent deformation. A record shall be made of any zero shift in the instrument during a test cycle

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where failure of the instrument to return to zero after loading may be an indication of yield along lines of principal stress other than the longitudinal axis of the gage.

(2) Crane boom. Conduct strain tests of the crane boom while loads of 294,000 pounds and 47,000 pounds are hoisted and lowered, at the radius specified in table I in the most stable direction. Boom out a rated load from the 20 feet operating radius to the tipping radius in the most stable direction. In addition, strain tests shall be conducted for those components not receiving their maximum stress in the most stable direction while tipping loads are hoisted, swung, and lowered at 20 feet and 60 feet operating radii in the least stable direction. Record maximum strains during each testing cycle on the strain gage locations described in (d). Swinging of loads over the side shall be through a cycle of 180 degrees and over the rear of each crawler corner to the other. Raise the 180-foot boom from ground level to 54 degrees from horizontal and take strain gage readings on all A-frame and gantry components.

(g) Records: Data shall be recorded on the following form:

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MAXIMUM WORKING STRESS

Contract Order No._____ Place_____
Front-End Attachment_____ Date_____
Recording Instruments_____ Recorder_____
Gage Type_____
Gage Factor_____
Gage Resistance_____

Gage No.	Component	Gage Location	Material[a]	Operation	(Recorded Strain Microinches) Test 1 (a) Zero Reading	(b) Test Reading	(b-a) Strain	Test 2 (a) Zero Reading	(b) Test Reading	(b-a) Strain	Test 3 (a) Zero Reading	(b) Test Reading	(b-a) Strain	Stress psi[c] Maximum[b]

a/ E = Modulus of elasticity.
b/ Where rosette gages are used, see manufacturer's instruction for exact strain.
c/ Stress = Actual Strain x E.

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(h) Maximum static and dynamic stress greater than 60 percent of yield strength shall constitute failure of this test. The yield strength and modulus of elasticity of the material of the structural member shall be obtained from "Engineering alloys" published by the Reinhold Publishing Corporation, from other recognized standards, or from certified laboratory test of the material concerned.

4.6.2.10 Maintainability. Maintenance operations listed in 3.9(a) through (g) shall be accomplished. Failure to accomplish the operation in the time specified with the tools specified shall constitute failure of this test.

4.6.2.11 Lifting capacity tests. Determine the lifting capacities of the crane by use of actual weights or load cells. Lift the loads specified in table I and swing 360 degrees. If a load cell is used, test loads shall be lifted over the front, rear and each side. Tipping or maximum capacity loads in the least stable direction less than those shown in table I shall constitute failure of this test. The overload indicating system shall be disconnected when lifting tipping loads in this test and 4.6.2.11.1.

4.6.11.1 Crane-capacity plate determination. Repeat the procedure specified in 4.6.2.11 for each additional 20 feet of boom length up to 180 feet. Plot 75 percent of the load values determined for 1-foot increments of minimum rated radius up to boom length radius for each 20-foot increment of boom length.

4.6.2.11.2 Craning. Using various rated loads with the basic boom perform craning operations for 50 hours at up to and including maximum radius for each load shown on the load chart. Inability to perform craning, inability to complete the test, malfunction, evidence of any damage, or failure of any crane component shall constitute failure of this test.

4.6.2.11.3 Overload indicating system. When required utilizing a 60 foot boom set at first for the minimum and then the maximum operating radius (see 3.15.1.4) perform the following test.

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1. Assemble a load of a size which will not activate the visual signal.
2. Lift the load 2 feet off the ground.
3. Increase the load by small increments and determine the load required to activate the audible signal.
4. Continue to increase the load and determine the load required to activate the visual signal.
5. Reduce the load and determine the load at which the audible signal ceases.
6. Continue to reduce the load to determine the load at which the visual signal stops.
7. Lower the load to the ground and repeat the above procedure for the 120 and 180 foot booms.
8. Failure of the overload indicating system and its components to meet the requirements of 3.15.1.4, inability to complete the test, or malfunction shall constitute failure of this test.

4.6.2.12 Rope strength. Measure the maximum load on all ropes when lifting a rated load at 15-foot radius and while raising a boom 180 feet long from the ground. A factor of safety less than specified in 3.18 shall constitute failure of this test.

4.6.2.13 Power and speed test. This test shall be conducted at an actual or simulated altitude of 5,000 feet. If the option is taken to simulate altitude conditions for test purposes, such simulation shall be accomplished in the following manner. In the vicinity of the engine, mount a container of sufficient capacity and inlet orifice size to assure a steady air pressure to the engine equal to 24.89 inches of mercury. Connect the outlet of the container to the air cleaner so that air is inducted into the engine in the normal manner through the air cleaner filter. Determine the intake air pressure with a U-tube manometer tapped into the container at a point where static air pressure is maintained in the container. The intake and exhaust systems shall be complete and unaltered. Exhaust back pressure shall not be reduced for this test.

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(a) Check clutches and brakes and adjust to manufacturer's specifications.
(b) By trial lifts, adjust a load to provide steady hoisting rate near full-load, governed engine speed. Establish a mark on the hoisting rope at a point near the primary hoist drum when the load is 3 feet clear of the ground. Establish a second mark on the rope when the load has been lifted to a point where the blocks are approximately 10 feet apart, or before interference occurs between the load and crane structure. Lower the load and measure the linear feet of rope between the two marks.
(c) Using a stopwatch accurate to plus or minus 0.2 second in 1 minute, time the interval required for the two marks on the hoisting rope to pass a selected point while the load is being lifted with the engine operating at maximum governed speed. At least three checks shall be made and the average time determined.
(d) Compute the hoist line speed in feet per minute and drum line power.
(e) Assemble a test load equal to the hoisting drum line power divided by the applicable line speed. Determine that the crane is capable of lifting this load, with a single hoist line, and simultaneously rotating the superstructure.
(f) Using a 60-foot boom, rest the tip on the ground and count the number of wraps of rope left on the boom hoist drum.

A hoist line speed greater than 200 feet per minute, a hoisting drum line power less than 6,000,000 ft-lb/min, inability to rotate the superstructure and simultaneously lift the single line test load, of less than three wraps of rope remaining on the boom hoist drum when the boom tip is resting on the ground shall constitute failure of this test.

4.6.2.14 Swing test. With the boom at 15 feet operating radius, rotate the superstructure at full throttle for 10 minutes in each direction. Check swing speed for 1 minute in each direction. Determine the temperature rise in the rollers. Load a dragline bucket to a gross weight of 18,000 pounds or an equivalent weight.

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With a 60-foot boom at not higher than a 50-degree angle make 90-degree swings as specified in 3.12.7 for 1 hour and determine temperature of swing clutches. A temperature rise in excess of 175° F. in the supporting balls or rollers, a temperature in excess of 600° F. in the swing clutches, swing speed not in accordance to 3.12.7, or fading of the swing clutches shall constitute failure of this test.

4.6.2.14.1 Clamshell and dragline tests. Perform maximum-capacity, heavy-duty clamshell and dragline operations in undisturbed heavy clay soil. The operation shall be performed with the clamshell and dragline bucket for at least 16 hours (8 hours for each bucket). Any evidence of malfunction, damage, or permanent deformation shall constitute failure if this test.

4.6.2.15 Power lowering test.

(a) Position the superstructure in the most stable position and assemble a load of 60,000 pounds with 1/2 the parts of load line for 120,000 pounds including the weight of the hook block.
(b) At 25 foot radius and with the brake unlocked, hoist the load 20 feet plus or minus 3 inches above the ground.
(c) Adjust the hoist drum brake to hold the load and apply the hoist drum brake until the brake will hold the load suspended.
(d) Prepare the crane for "power-down" operation on the load line.
(e) Power-lower the load with the engine speed adjusted to override the applied brake adjusted as specified in (c) above.
(f) As soon as the load reaches 1 foot above the ground, immediately repeat operations (b) and (e). Repeat until 30 cycles have been completed.
(g) Check the temperatures of all brake linings and drums employed in this operation with at least four thermocouples imbedded within each lining near the working face and uniformly spaced around the perimeter or a contact pyrometer.

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(h) Hoist to a height of 5 feet above the ground a load that will develop a line pull in pounds equal to the maximum single line pull developed by maximum rating and divided by recommended parts of reeving. Apply the hoist drum brake to hold the load. Stop the engine and jam-lock a meshing set of gears between the engine and the power-down clutch. Release the hoist drum brake and let the load be supported. Temperature of brake linings in excess of 600° F. or the maximum recommended by the manufacturer of the lining for the application, whichever is less, inability to complete 30 cycles, or inability to hold the load shall constitute failure of this test.

(i) Check power-down and hoist line drum speed at rated engine speed, using a tachometer on the drum shaft. Power-down line speed not in accordance with 3.11.1 shall constitute failure of this test.

4.6.2.16 Load holding test. Suspend on a single line a load that will develop the single-line pull resulting from maximum capacity; stop the engine; jam-lock a meshing pair of gears; by carefully wedging a brass or other soft rod between the gears; positively apply the primary hoist drum clutch and release the brake holding the load. As the brake is released, examine the clutch for slippage. Repeat this procedure using the secondary hoist drum clutch. Suspend the load used for the previous procedure; stop the engine; positively engage both the secondary and primary drum clutches; disengage the master clutch and lower the load on the brake backing down the hoisting machinery. Determine ease of control operation, smoothness of load lowering, and machinery operation. Binding, chattering of machinery, raising of the boom or slipping at the clutches shall constitute failure of this test.

4.6.2.17 Propelling. Self-propel the crane with basic boom and hook block, for 20 miles at an average speed of not less than 0.4 mile per hour over smooth, firm, dry ground, including 2 miles up and down grades of not less than 20 percent and not less than 1 mile along 10 percent side slopes. Steer the crane to the right and left to prove the steering capability. Any breakage, inability to complete the test, inability to steer, leakage of fluid, or malfunction shall constitute failure of this test.

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4.6.2.18 Torque-converter stall test. Stall the torque-converter at full throttle of the crane engine for not less than 36 seconds at an ambient temperature not less than 30° F. Then self-propel the crane for a period of 15 minutes. Oil temperature exceeding 250° F. from an initial temperature of not less than plus 130° F. in an ambient temperature of not less than 30° F. or inability of the torque-converter oil to return to a stabilized operating temperature within 15 minutes shall constitute failure of this test.

4.6.2.18.1 Hoist and travel clutch stall. Stall the torque converter using the hoist clutch and repeat using the travel clutch. Failure of either clutch to stall the torque converter with the engine throttle at maximum open position shall constitute failure of this test.

4.6.2.19 Sectionalization test. Disassemble the crane for shipment. Then reassemble and operate for 5 minutes. Inability to disassemble the crane in 96 manhours without any special tools, cutting or welding required, any section which exceeds the limiting dimensions specified in figure 1, or inability to reassemble in 96 manhours and operate the crane shall constitute failure of this test. The counterweight shall be removed or replaced utilizing only those lifting provisions provided on the crane.

4.6.2.20 Lifting attachments.

a. Lift the crane and hold in suspension in its normal transport position using slings that converge not more than 24 feet above the lowest extremity of the crane. Determine the forward angles of application for each attachment. Measure the clearance between each sling and the crane. Measure eye openings and clearance dimensions of the attachments.
b. Restrain the crane by anchoring the main frame and subject each attachment to a force of 2-1/2 times the load it carried when initially suspended. This force shall be applied in the direction as determined in a. above. An alternate method of test is to lift the crane as in a. above and add weights to the main frame until the force

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is 2-1/2 times the initial force on each attachment. Hold the crane under either of the above loading conditions for at least 90 seconds.

4.6.2.20.1 Failure criteria. Weld failure or permanent deformation of the crane at points other than the lifting attachments, or failure to meet the requirements of MIL-STD-209, shall constitute failure of this test.

4.6.2.21 Tiedown attachments. With the crane anchored by means other than the attachment being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedowns.

4.6.2.21.1 Failure criteria. Breakage or failure to meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.6.2.21.2 Railroad humping test. Anchor the crane on a domestic service railway flatcar (blocking is permitted). Hump the car, with the crane tied down, at speeds of not less than 4, 6, and 8 mph against 5 empty boxcars with their brakes set. The crane shall face in the direction of the 5 cars. Repeat the procedure at not less than 8 mph with the rear of the crane in the direction of the 5 cars. The crane may be reversed by either reversing the direction of the flatcar (preferred) or by reloading the crane in the proper direction on the flatcar. Once the test has begun, there will be no readjustment of load, nor any reconditioning of the blocks, checking material or tiedowns except as necessary to reverse the direction of the crane. After each impact, the crane shall be inspected for permanent deformation to its components, or other evidence of possible failure. After completion of the railroad impact test, an operational checkout of the crane will be conducted.

4.6.2.21.2.1 Failure criteria. Weld failure, or permanent deformation to the components of the crane or damage to any tiedown cable or stopping of a test due to failure and damage to the crane, or failure of the crane to operate after the test shall constitute failure of the test.

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4.6.2.22 Antirotation test.

(a) Install the antirotation rods and tie the crane down to prevent tipping. Release the swing brake and the swing lock of the crane. Reeve a line to a point on one end of the superstructure to exert a moment in a horizontal plane both clockwise and counterclockwise. The moment shall be not less than 2 times the weight of the superstructure (without boom) times the distance from the center of gravity to the center of rotation.
(b) Repeat the procedure using one-half the force applied in (a) above with the swing lock engaged and the attirotation rods and swing brake released.
(c) Repeat the procedure with a moment of not less than 0.30 of the force applied in (b) above with the swing brake set and the antirotation rods and swing lock released.

Breakage or distortion of the rods, brackets, attaching members, or welds of inability of either the antirotation rods, swing lock or brake independently to hold the superstructure in place shall constitute failure of this test.

4.6.2.23 Electromagnetic interference. The preproduction model shall be tested to determine conformance with 3.20. The supplier shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6.2.24 Operating temperature. Thermocouples shall be installed in the crane as follows:

(a) Engine coolant temperature: In top tank radiator connection.
(b) Engine lubricating oil: In the oil gallery.
(c) Battery temperature: In the center cell of each battery.
(d) Torque converter oil: In torque converter sump.

If required, due to limited space, operating-temperature tests may be performed using the superstructure only.

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4.6.2.24.1 High-temperature operation. Subject the crane to an ambient temperature of 125° F. until the temperature of the crane has stabilized. Start the engine and operate it at maximum governed speed for 1 hour. During this hour, individually engage and disangage the clutches. Actuate the brake controls. Move the superstructure clockwise and counterclockwise within the limitations of available space. Stop the engine at the end of 1 hour; restart the engine after 5 minutes and operate for 10 minutes without load. Failure of the engine to start and operate within the limits specified or failure of any component to function as specified herein shall constitute failure of this test.

4.6.2.24.2 Low-temperature operation. Subject the crane to an ambient temperature of minus 25° F. until the temperature of the crane has stabilized at minus 25° F. including the temperature of the battery electrolyte. Start the engine with its batteries and run to determine functioning of the crane. Individually engage and disengage supersturcture clutches under maximum governed engine speed. Actuate brake controls. Move supersturcture clockwise and counterclockwise within limitations of available space. Failure of the engine to start and operate or failure of any component to function correctly shall constitute failure of this test.

4.6.2.25 Cold starting and running. The crane when winterized shall be tested in accordance with test method No. 1. One or more of the following shall constitute failure of this test:

(a) Diesel engine does not start within 5 minutes after 4 hours of standby heat.
(b) Engine does not run smoothly within 15 minutes after starting without unnatural or continued control manipulation and without the use of primers and priming fluids.
(c) Engine oil sump temperature lower than plus 10° F. prior to engine starting.
(d) Battery temperature lower than plus 10° F. prior to engine starting.
(e) Engine coolant temperature at heater inlet lower than plus 50° F. prior to engine starting.

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(f) Battery compartment temperature less than plus 50° F. subsequent to engine start.
(g) Cab temperature less than plus 35° F. at operator's foot pedals within 1 hour after starting engine.
(h) 75 percent of each window not frostfree within 1 hour after starting engine.
(i) Lubricating oil temperature not stabilized in the engine lubricating oil gallery within the range of 130° F. to 230° F. within 1 hour after starting and remain within this range throughout the 1-hour test period.
(j) Coolant temperature not stabilized in the engine-cooling jacket discharge pipe within the range of 140° F. to 210° F. within 1 hour after starting and remain within this range throughout the 1-hour test period.
(k) Cab temperature at operator's foot pedal after cab heater has operated 1 hour could not be maintained at plus 35° F. or warmer.
(l) Instruments and gages do not function properly.
(m) Thermal damage to batteries.
(n) Battery charging system does not show positive charge.
(o) While the operator is wearing arctic mittens and boots:
    (1) Power train cannot be engaged to the crawler treads and drums.
    (2) All clutches, brakes, shafts, drums, and hydraulic cylinders cannot be operated through two movement cycles.
    (3) Windshield wiper(s) cannot be electrically and manually operated.
    (4) All cab doors, engine housing doors, and winterfronts cannot be operated, latched, and unlocked.
    (5) Operator cannot safely mount to the operator's position in the crane and cannot safely dismount from the operator's position.
    (6) Maintenance checks cannot be performed.

4.6.2.26 Sound level measurement.

4.6.2.26.1 Operator's station. Measure the sound level at the operator's station in accordance with MIL-STD-1474 for steady-state noise, stationary equipment, at the operating condition of governed engine speed lifting 2/3 of a rated load.

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4.6.2.26.2 Representative duty position. Measure the sound level at eight evenly space points, 68 inches above ground level, on a circle of 25 foot radius from the geometric center of the equipment in accordance with MIL-STD-1474 for steady-state noise. The operating condition shall be governed engine speed lifting 2/3 of a rated load.

4.6.2.26.3 Exterior spectator. Measure the exterior sound level in accordance with MIL-STD-1474 for exterior noise.

4.6.2.26.4 Non-conformance to the requirements in 3.8.1 shall constitute failure of this test.

4.6.2.27 Production-run test. Each crane shall be operated for at least 2 hours as a dragline. Any evidence of nonconformance to 3.15 shall constitute failure of this test.

4.6.2.27.1 Lights and horn. Operate each of the switches controlling all lights and the horn button for not less than five times. Failure of the switches to operate, lights to burn clearly or horn to function properly shall constitute failure of this test.

4.6.2.27.2 Windshield wiper. Turn on windshield wipers and observe that the blades operate making full sweeps rhythmically. Failure of the windshield wiper to operate shall constitute failure of this test.

4.6.2.27.3 Overload system. Perform the test specified in 4.6.2.11.3 using the basic boom and the minimum operating radius. Nonconformance to 3.15.1.4 shall constitute failure of this test.

4.6.2.27.4 Engine production test. Each engine shall be tested in accordance with Test Method Series 4000, MIL-STD-1400 except that the engine manufacturer standard commercial run-in may be used, provided it equals or exceeds Test Method Series 4000, MIL-STD-1400.

4.6.2.28 Maintenance package. During initial production testing, maintenance support elements such as manuals, when furnished, lubrication charts, repair parts and special tools (see 6.7 and 6.8) shall be evaluated. Failure of the crane to be properly maintained utilizing the support elements shall constitute failure of this test and shall require revision of the document(s).

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4.6.2.29 Boom stop.

(a) Calculate the load imposed on the boom stop and attaching points when bending a boom over the stops. Subject the boom stop to this load. The basic boom shall be lowered until the boom point touches the ground; the raise the point until the boom operating radius is at 13 feet. Repeat this cycle for a total of not less than 12 times, all at maximum boom hoist speed.

(b) Failure of boom stop to withstand force required for completely bending the boom, failure of boom stop to stop the basic boom at 13 foot operating radius, any noticeable binding or chattering of the boom stop when raising boom, or failure of the boom hoist cutoff to operate shall constitute failure of this test.

4.6.2.30 Crane hook. After completion of the maximum working stress test, the crane hook shall be given a wet or dry ferromagnetic particle test for suface or subsurface cracks or discontinuities. Either method of test used shall show clearly visible particles defining existing discontinuities. Any indication of cracks or discontinuities shall constitute failure of this test.

4.6.2.31 Rain. Subject the crane to a simulated rainfall of not less than 4 inches per hour as measured by a US Weather Bureau type gage. The simulated rainfall shall fall on the crane, impinging with a force equivalent to a 30 mile an hour wind. The simulated rainfall shall disperse uniformly over the surfaces of the crane for not less than 30 minutes on each surface of the crane. The engine shall be started (without removing accumulated water) 26 minutes after beginning the rain test, and idle for the remainder of the test for each of the four sides of the crane. At the completion of the above (within 30 minutes), the crane shall be operated and lift and swing a rated load. Inability of the engine to start and idle, or inability to perform craning operations as specified herein, shall constitute a failure of this test.

4.6.2.32 Fording. Water fording capabilities shall be ascertained as follows for the preproduction model. (a) Bring crane to operating

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temperature. (b) Operate a minimum of 2 hours with not less than 12-inch deep water; immediately thereafter, operate on shore in high gear for 2 hours. (c) Sample oil from engine and all drive components. (d) Repeat above four times except the crane shall be stopped a minimum of four times. The initial production test items shall be subjected only to para 4.6.2.32(b) above. Inability to complete this test or nonconformance to 3.10.2.1 shall constitute failure of the test.

4.7 Inspection comparison. The Government may select cranes at any time during the contract production period and subject these cranes to the examination and tests specified in 4.4, to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production cranes. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of these tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison crane to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison crane(s) and shall be cause for refusal by the Government to continue acceptance of production cranes until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on cranes previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all cranes accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

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4.8 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined and tested in accordance with the Quality Assurance Provisions of MIL-C-3580.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Each complete crane shall be preserved, packaged, packed and marked in accordance with MIL-C-3580. The level of preservation and packaging shall be level A, B, or C and the level of packing shall be level A, B, or C as specified (see 6.2).

6. NOTES

6.1 Intended use. The cranes, together with the various items of front-end operating attachments, are intended for use in handling materials, lifting and transporting loads short distances, and soil stripping.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of preproduction model cranes, time frame required for submission, and where test(s) shall be performed (see 3.2.1).
(c) When the crane shall be winterized (see 3.2.1.1).
(d) When initial production testing is required, number of initial production cranes to be furnished (see 3.3).
(e) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(f) When a priming system shall not be furnished (see 3.13.5).
(g) Front-end attachments to be furnished (see 3.15).

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(h) When crane boom length shall be other than 60 feet (see 3.15.1).
(i) When overload system shall be furnished (see 3.15.1.4).
(j) Color required (see 3.29).
(k) Level of preservation and packaging and level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production cranes from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish cranes conforming to this specification.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the crane covered by this specification as contained in the Federal Supply Catalog, DOD Section, Identification List C9100-IL for FSC Group 91.

6.5 The contracting officer should arrange to furnish the Electrical Warning Decal Set, FSN 7690-267-5778 and provide mounting instructions as required in paragraph 3.27.

6.6 Information figures. Figures 2 and 3 show a type of labyrinth seal which has been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other labyrinth seal shapes which conform to this specification.

6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each crane.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of cranes. A suggested paragraph is as follows:

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"Shipment of cranes shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

Custodians:

Army - ME
Air Force - 84

Preparing activity:

Army - ME

Project No. 3810-0051

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WINTERIZATION TEST (MINUS 50° F.)

TEST METHOD NO. 1

1. TEST APPARATUS:

   (a) Cold chamber.
   (b) Arctic clothing.
   (c) Thermocouples.
   (d) Battery hydrometer.
   (e) Wattmeter.
   (f) Voltmeter.
   (g) Recording instruments (temperature readings), electrical.

2. TEST PROCEDURE:

   (a) Service the crane with arctic antifreeze conforming to MIL-C-11755 and with arctic lubricants.
   (b) Check the specific gravity of each battery. Specific gravity shall be not lower than 1.275 before beginning of cooldown.
   (c) During the 4-hour standby heating and starting period, close and latch the housing panels. All operations incident to closing and covering up the engine shall be performed by one operator wearing arctic mittens.
   (d) All operations incident to starting and priming the engine shall be performed by one man wearing arctic mittens.
   (e) Subject the crane to an ambient temperature of minus 50° F. until all parts of the crane have stabilized at minus 50° F. Perform the test after stabilization is assured by temperature readings of minus 50° F. from thermocouples located as follows:
      (1) In the two center cells of each battery between the plates and at middepth of the electrolyte.
      (2) In the volumetric center of any hydraulic oil tank of more than 25-gallon capacity.
      (3) In the engine lubricating oil gallery (at downstream end).
      (4) In the engine coolant in the engine-cooling-jacket discharge pipe.
      (5) In the chamber at a horizontal angle of 45 degrees, from

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the sides and ends of the crane and not less than 3 feet from each corner or the crane at engine-crankshaft level.

(6) In the operator's compartment 6 inches above the floor boards and between the operator's foot pedals.

(f) Maintain minus 50° F. ambient temperature throughout the test.

(g) Perform preheating by interconnecting the crane to outside 115-volt power source using the 25-foot cable supplied with the crane for 4 hours. Start coolant-type cab heaters not more than 15 minutes after the engine has been started.

(h) Substitute or supplementary batteries or other electrical cranking energy shall not be used.

(i) Glow plugs or other accessories working off of the cranking batteries may be used in the minus 50° F. start. Use only the priming aids specified in 3.13.5.

(j) Start the diesel engine within 5 minutes after initial 4 hours of standby heat.

(k) Operate the engine until:

(1) The lubricating oil temperature is stabilized in lubricating-oil gallery within the range of 130° F. to 230° F.

(2) The coolant temperature is stabilized in the engine cooling jacket discharge pipe within the range of 140° F. to 210° F.

(l) Run the engine for 1 hour after these conditions have been met.

(m) Cycle the engine periodically between idling and full load governed speed during the 1-hour period of engine operation specified in (1) above.

(n) Record at 10 minute intervals temperatures during 4 hour standby heating period, immediately after engine starting, during engine warmup, and during temperature stabilization and 1-hour operating period, including but not limited to the following:

(1) Batteries.

(2) Hydraulic tank.

(3) Engine oil.

(4) Engine coolant.

(5) Ambient temperature.

(6) Operator's compartment 6 inches above the floor at the foot pedal.

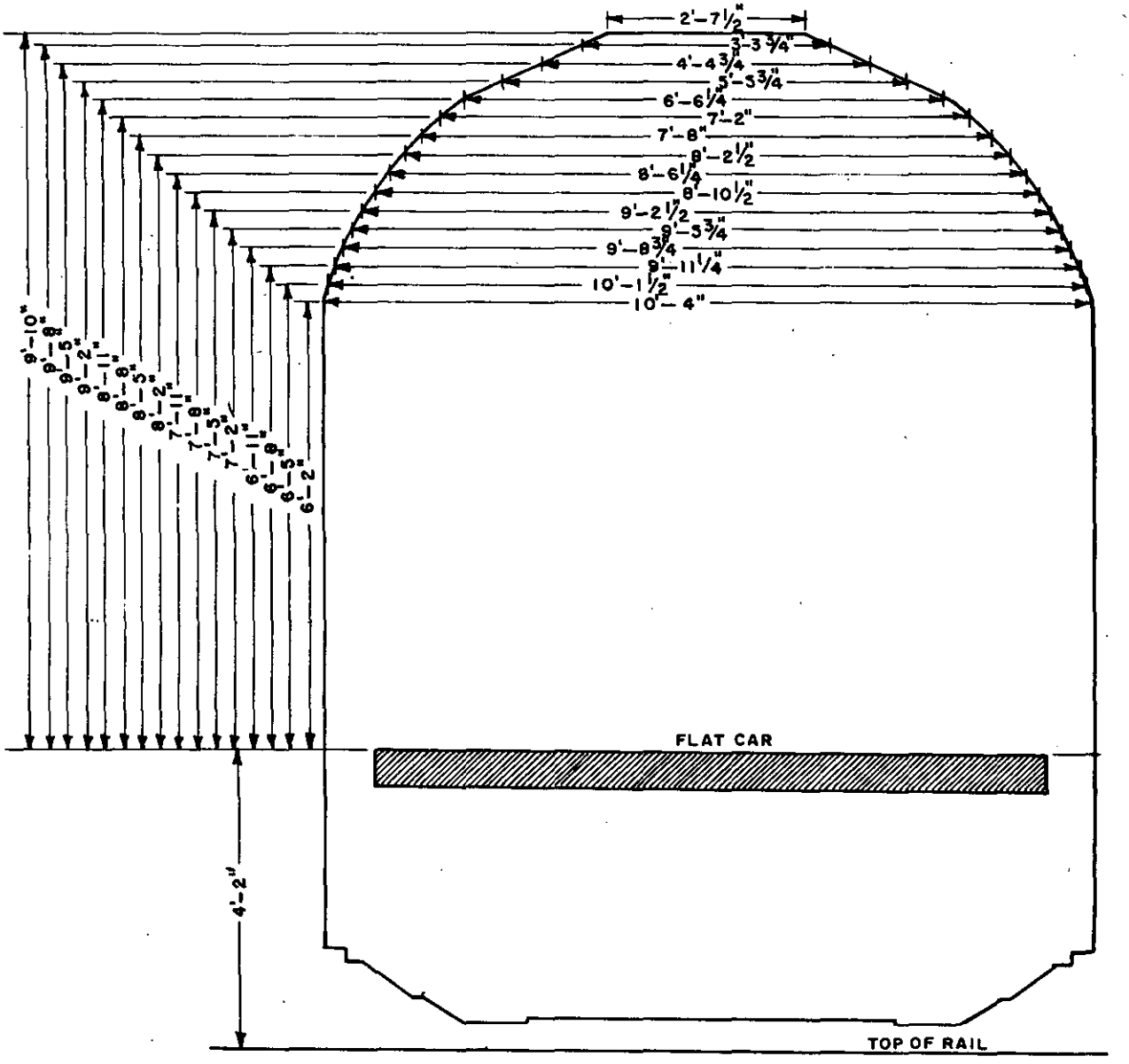
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(7) Converter oil at outlet.
(8) Transmission oil.

(o) During 1-hour operation and while wearing arctic clothing, an operator shall perform each of the following:
(1) Engage and disengage power train.
(2) Operate through two complete movement cycles all clutches, brakes, shafts, drums, cylinders, and hydraulic components.
(3) Operate window wipers.
(4) Operate all cab doors, engine housing doors, sidehoods, and winterfronts by opening, closing, latching, and unlocking each of these items.
(5) Mount to and dismount from the operator's position on the crane.

(p) Upon completion of the 1-hour period, shut down the engine, and while the crane is still in the chamber at minus 50° F., and the operator is wearing arctic mittens, he shall perform each of the following maintenance checks and operations:
(1) Restart the engine by using outside power through the battery-charging receptacle. Disconnect cable to outside power and then shut down the engine.
(2) Check oil level on all dipsticks.

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ITEM LENGTH - 32'-10"

FIGURE 1. BERNE INTERNATIONAL
(CLEARANCE DIAGRAM)

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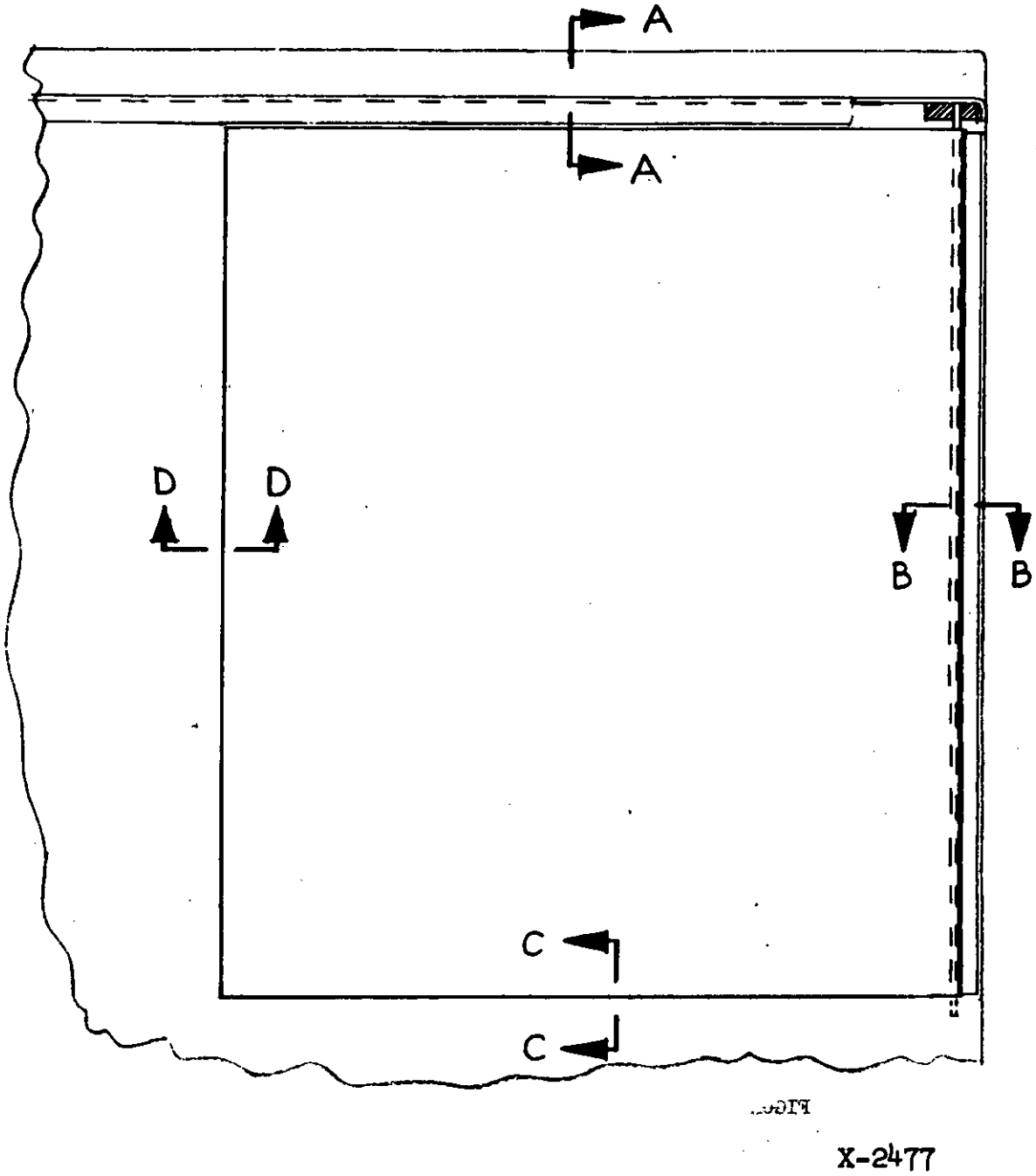


FIGURE 2. LABYRINTH SEAL TYPICAL VIEW

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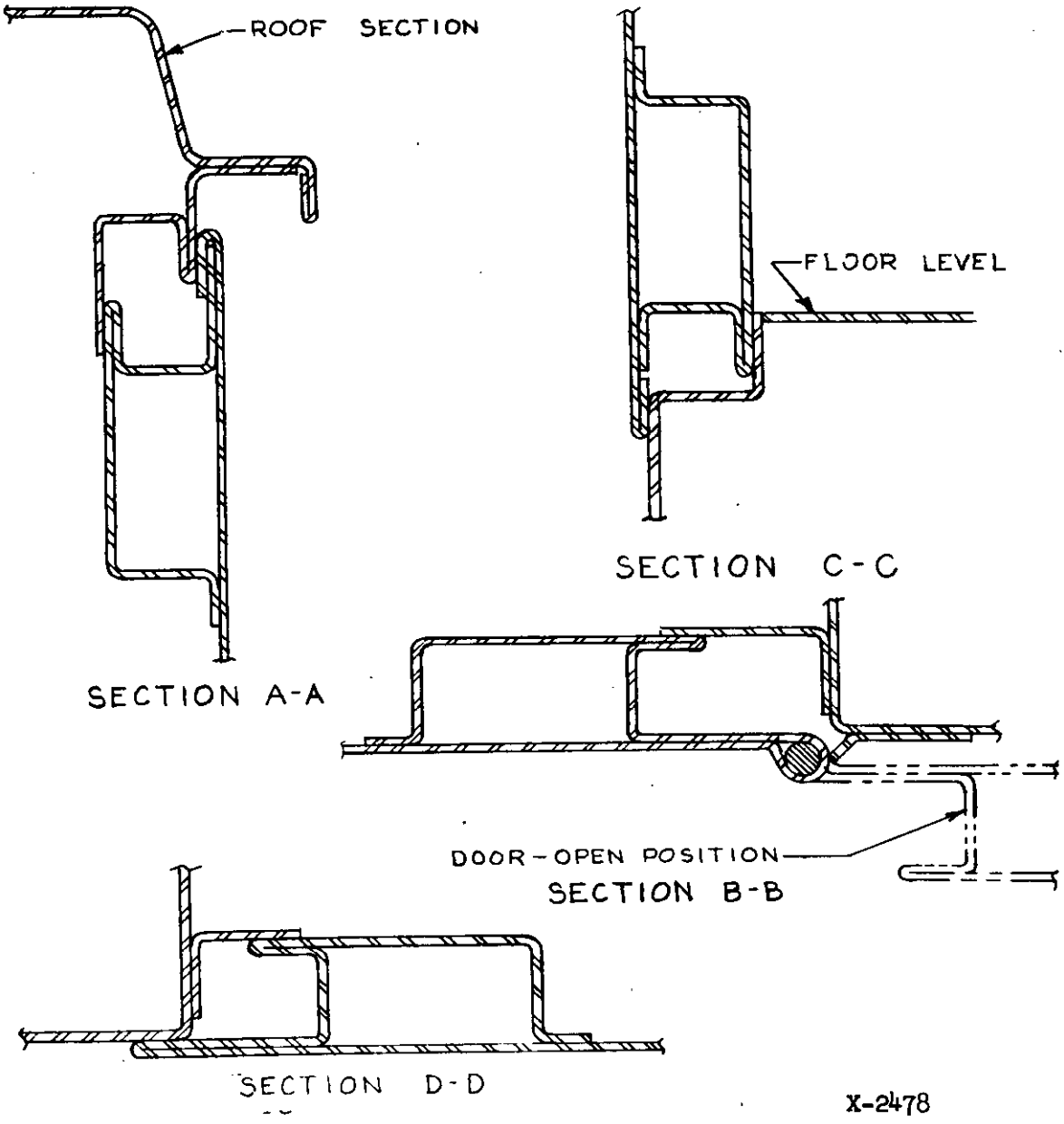


FIGURE 3. **LABYRINTH SEAL TYPICAL VIEW SECTIONAL VIEWS**

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-52326B - CRANE, CRAWLER MOUNTED: 60-TON AT 25-FOOT RADIUS, DIESEL-ENGINE-DRIVEN

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

42,615—Ft Belvoir

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U. S. ARMY MOBILITY EQUIPMENT
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